

The Quantum Cost of Unit-Normalized Spectral Shifting

Abstract

We determine the oracle cost of converting a block encoding of a positive semidefinite contraction H into a reusable encoding of $I - H$ with normalization exactly one. For spectra in $[\delta, \rho\delta] \cup [c, 1]$, with fixed $\rho > 1$ and $0 < c \leq 1$, globally nonnegative polynomial approximation gives explicit thresholds G_r . At error $K\delta$ with fixed $K > 0$, the optimal query count is $\Theta(\delta^{-1+1/(2\ell)})$ for $\ell = \min\{r : G_r \leq K\} \geq 1$, including threshold equality; we also determine the coarse tier. An even-polynomial threshold plateau yields a sharp $\Theta(\delta^{-5/6})$ versus $\Theta(\delta^{-1/2})$ separation from unrestricted conversion at a concrete accuracy, while odd transforms cost $\Theta(\delta^{-1} \log(1/\delta))$. Uniformly for $0 < K \leq \delta^\beta$ with fixed $\beta > 0$, the unrestricted cost is $\Theta(\delta^{-1} \log(1/K))$. In the intermediate joint regime we quantify a remaining multiplicative gap. The lower bounds cover general coherent oracle circuits, and the upper bounds preserve global contractivity. An exactly central sparse linear program realizes matched normal-matrix versus factor-access bounds for the dual Newton state and transfers the complement hierarchy, with a zero-query coarse exception. These are analog oracle separations, not lower bounds for solving the linear program.

1 Introduction

A positive semidefinite contraction H and its complement $I - H$ have equally simple classical descriptions. Their quantum access costs can be very different. A block encoding supplies H as a designated block of a unitary. A coherent linear combination with the identity immediately supplies $(I - H)/2$, but an algorithm that requires $I - H$ itself cannot discard the factor of two. Near a small eigenvalue of H , the desired complement eigenvalue approaches the boundary of the unit disk. Global unitarity then restricts how accurately a finite-query circuit can follow that eigenvalue on a whole interval.

This normalization issue already appears explicitly in Orsucci and Dunjko’s positive-definite linear-system algorithms [16, Section 4.3]. They require a normalized encoding of $I - \eta A$, explain the limitation of generic linear-combination constructions, and give specialized constructions under stronger access assumptions. We quantify the conversion cost from a plain block encoding when the spectrum has a shrinking low interval and a separated high band. The question is about a reusable coherent operator, so learning an instance and preparing one output state is a different task.

1.1 Problem and results

Fix a low-band ratio $\rho > 1$ and a high-band endpoint $0 < c \leq 1$. Let $\delta > 0$ tend to zero with $\rho\delta < c$, and suppose

$$\text{spec}(H) \subseteq [\delta, \rho\delta] \cup [c, 1].$$

We seek an output unitary whose designated block approximates $I - H$ in operator norm to error η , with normalization exactly one. Let $Q(\delta, \eta; \rho, c)$ be the minimum worst-case number of controlled calls to the input block encoding or its inverse. The converter must work for every matrix and every unitary completion consistent with the promise; its gates may depend on the public parameters, but

not on the unknown matrix. The count is the cost of applying the output conversion circuit; reusing it requires its oracle calls again. Bounded-query adaptive algorithms are included by deferring measurements. Section 2 states the complete contract.

For fixed relative accuracy $K > 0$, meaning $\eta = K\delta$, the optimal query exponent changes discretely with K . The transition values are

$$G_r(\rho) = \inf_{\substack{P \in \mathbb{R}[y], \deg P \leq 2r \\ P \geq 0 \text{ on } \mathbb{R}}} \max_{1 \leq y \leq \rho} |P(y) - y|, \quad r = 0, 1, 2, \dots$$

Our first main result computes these thresholds explicitly and proves the full query staircase. If $\ell = \min\{r : G_r \leq K\} \geq 1$, then

$$Q(\delta, K\delta; \rho, c) = \Theta(\delta^{-1+1/(2\ell)}).$$

The constants here can depend on the fixed parameters. Every equality $K = G_\ell$ belongs to the cheaper tier. The coarse tier $K \geq G_0 = (\rho - 1)/2$ costs $\Theta(\log(1/\delta))$ when $c < 1$, and $\Theta(1)$ when the high band is the singleton $\{1\}$. Theorems 4.1 and 4.2 give the precise statements. Figure 1 illustrates the exponent transitions. In contrast to a continuum, suitable known two-point spectra admit an exact two-query conversion (Proposition 3.1). No finite-query converter is exact on a nonempty interval at normalization below two (Proposition 3.2).

The second result measures the cost of a parity restriction. A single real definite-parity quantum singular value transformation (QSVT) has a bounded even or odd polynomial as its transformed block. The even class has its own optimal staircase, governed by the same approximation problem with P also required to be even; denote its thresholds by F_r . We prove the exact plateau

$$F_1(\rho) = F_2(\rho) = \frac{(\rho - 1)^2}{8(\rho + 1)}.$$

Thus the even staircase skips the exponent-3/4 tier. At $\rho = 2$ and $\eta = \delta/32$, a globally positive degree-six witness gives matching orders $\Theta(\delta^{-5/6})$ for the even class and $\Theta(\delta^{-1/2})$ for unrestricted coherent conversion. The witness bounds $F_3(2)$ from above; it does not compute it. Odd transforms have the matched order $\Theta(\delta^{-1} \log(1/\delta))$ at every fixed $K > 0$ (Theorem 4.11 and Proposition 4.12). These are restrictions on a single definite-parity transform, not on arbitrary compositions of QSVT circuits. Generalized quantum signal processing (GQSP) supplies the unrestricted upper bounds without a normalization loss.

Third, we control joint limits in δ and accuracy. For every fixed $\beta > 0$, uniformly over $0 < K \leq \delta^\beta$,

$$Q(\delta, K\delta; \rho, c) = \Theta_{\rho, \beta}(\delta^{-1} \log(1/K)).$$

For the full spectral interval $[\delta, 1]$, the corresponding order is $\Theta_\beta(\delta^{-1} \log(1/\eta))$ when $0 < \eta \leq \delta^{1+\beta}$ (Theorem 5.3). In the intermediate regime $K \rightarrow 0$ with $\log(1/K) = o(\log(1/\delta))$, let $r = \min\{j : G_j \leq K\}$. Uniform lower and upper bounds have the same power $\delta^{-1+1/(2r)}$ and prefactors $r(G_{r-1} - K)^{1/r}$ and r^3 , respectively. If $K \leq (1 - \gamma)G_{r-1}$ for fixed $\gamma \in (0, 1)$, their ratio is at most $O(r^2) = O(\log(1/K)^2)$, including $K = G_r$ for large r . Close to the upper tier boundary the displayed margin factor remains necessary in our bound. We do not determine a uniform multiplicative optimum in that regime (Corollary 5.8).

Finally, we realize the access distinction in a three-variable, two-equality sparse linear program (LP) at an exactly central primal-dual point. Its normal matrix is $H_t = A_t A_t^T$, with hidden $t \in [\delta, \rho\delta]$ and condition number $\Theta(\delta^{-1})$. Preparing the normalized *dual* Newton direction to a fixed trace error costs $\Theta(\delta^{-1})$ queries from the specified normal-matrix oracle and $\Theta(\delta^{-1/2})$ from the specified

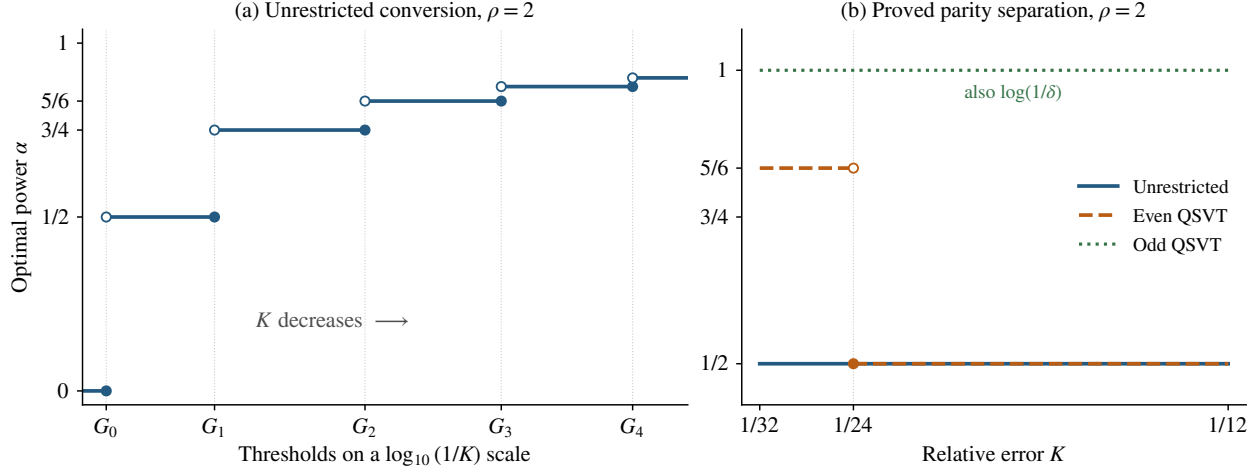


Figure 1: Optimal query powers for fixed relative accuracy, with $\rho = 2$. Left: the unrestricted exponent α in $Q = \delta^{-\alpha+o(1)}$ as K decreases. At a threshold, a filled marker selects the cheaper tier; an open marker excludes the more expensive one. The coarse exponent zero includes a logarithmic cost for $c < 1$. Right: the proved parity comparison on $1/32 \leq K \leq 1/12$. The even plateau $F_1 = F_2 = 1/24$ yields a jump from exponent $1/2$ to $5/6$; the odd class also has a logarithmic factor. No value of F_3 is inferred from the plotted comparison. Threshold locations on the left are numerical evaluations of the exact formula in Theorem 4.2; all complexity statements follow from proofs.

rectangular-factor oracle, even after counting access to the right-side preparation unitary. Separately, a matrix-only reusable complement compiler inherits every positive tier and the joint high-accuracy law, whereas factor access gives an exact two-query complement. Public spectral projectors make the LP compiler's coarse tier zero-query (Theorems 6.2 and 6.3). The example concerns these analog access contracts: its primal predictor is public, and digital entry access reveals the hidden parameter. It is not a lower bound for solving the LP or for an entire quantum interior-point method.

1.2 Related work and contribution scope

Quantum signal processing and QSVT provide the bounded-polynomial implementation tools used throughout this paper. Gilyén, Su, Low, and Wiebe [9] establish general singular-value transformations, bounded sign approximation, uniform amplification, and pairwise eigenvalue-transformation lower bounds. Low and Chuang [14] develop uniform spectral amplification and its normalization accounting. Motlagh and Wiebe's GQSP [15] removes the usual definite-parity restriction through bounded unit-circle polynomial synthesis. We use these implementation results, including a Laurent-polynomial reduction that works for arbitrary input completions. Neither parity removal nor the generic amplification upper bound is a contribution here. Classical phase computation and finite-precision circuit synthesis are separate resource questions; Haah [10] gives a rigorous algorithmic treatment of such decomposition costs. Our complexity measure counts oracle queries and allows exact oracle-independent gates.

The relationship between analog quantum queries and trigonometric polynomials is established prior work. Bessen [1] uses it to compare real-valued query models, and Laneve [12] characterizes QSP using state conversion and adversary feasibility. Our scalar-oracle reduction is an instance of this general polynomial method. The additional step is to combine global unitarity with a shrinking correctness interval: rescaled low-degree Taylor limits of the output defect must be nonnegative on

the entire real line. This identifies the specific extremal problem G_r and produces lower bounds for general coherent circuits, including circuits outside a single QSP sequence. The pairwise square-root obstruction alone does not give the higher tiers.

Constrained polynomial approximation also has a long history. Taylor [19] studies restricted ranges, Lewis [13] treats convex constraints, and Campos-Pinto, Charles, and Després [4] develop algorithms for positive polynomial approximation. Fawzi, Saunderson, and Parrilo [8, Sections 3.1–3.2 and Appendix A, Lemma 1] study globally nonnegative interpolation of affine targets at finite nodes and use Chebyshev supporting tangents. Our threshold problem instead minimizes uniform error on $[1, \rho]$ under global nonnegativity on $\mathbb{R}$ and determines shrinking-window query costs. Its solution uses the classical exterior Chebyshev inequality, as stated for example by Bos, Ma’u, and Waldron [2, Proposition 2.1]; our quantitative lower bounds also use classical Fejér–Riesz factorization. The odd upper bound uses established sign approximants from QSVT, with approximation-theory antecedents in Eremenko and Yuditskii [6, 7]. The gap-times-log-error behavior of sign approximation is therefore not new. Here it yields a matched comparison for the shrinking-window shift task, with an elementary lower bound that uses correctness only on the positive low interval.

Sarkar and Yoder [17] prove density results for QSP-related polynomial families with range constraints both inside and outside the approximation interval, under necessary endpoint matching conditions. They also study step-function approximation. Their qualitative approximation results provide further context for global constraints; our question asks for quantitative degree thresholds and coherent-query costs as a correctness interval shrinks.

A particularly close recent comparison is Dong, Larsen, Lin, and Sarkar [5]. They formulate prescribed-parity QSP design as minimax fitting on a subset subject to a global modulus bound, study Remez methods with active constraints, and introduce nonlinear Fourier retraction to restore feasibility without increasing the degree. Their work makes clear why satisfaction of bounds at sampled points is insufficient, especially near the boundary of the feasible set. We address an analytic asymptotic question for one target and spectral promise: exact limiting thresholds, optimal all-circuit query powers, and constructions that attain the threshold error itself. The global contractivity and contact preservation of our constructions are proved on continuous domains; numerical plots are only illustrations. We do not propose a general minimax solver or a phase-synthesis algorithm.

The closest direct motivation remains Orsucci and Dunjko [16]. Their Section 4.3 already formulates normalized $I - \eta A$ construction, invokes QSVT lower bounds against generic amplification, and gives sparse-value constructions for diagonally dominant matrices and local positive semidefinite sums. Their factor-based linear-system algorithms also explain square-root condition-number improvements and the associated support-overlap requirement. Our plain-block analysis does not resolve their separate general sparse-value construction question. Factor and sum-of-squares access are likewise central in King et al.’s spectral amplification [11]. Somma and de Wolf [18] prove joint lower bounds for guided ground-state energy estimation and state preparation, including sum-of-squares access, with guide-overlap and dimension assumptions. Those output tasks and hypotheses differ from reusable complement conversion; their bounds are not transferred here. The LP contribution is the explicit exact-central realization, its fully specified oracles, and the matched dual-state and compiler comparisons. Its state upper bounds use standard amplitude estimation [3].

Relative to these works, our contributions are the explicit globally nonnegative shift thresholds and their complete fixed-accuracy coherent-query staircase; the even-threshold plateau and sharp concrete parity separation; uniform continuum lower bounds matching high-accuracy amplification and quantifying the remaining intermediate gap; and the exact-central LP realization under matched access contracts. To our knowledge, this combination of exact thresholds, equality-attaining constructions, and optimal shift-query exponents has not appeared in the cited literature. This

claim concerns the stated problem and theorems, not the underlying synthesis, approximation, or factor-access techniques.

1.3 Proof structure and roadmap

Section 2 connects block encodings to bounded polynomials and represents scalar-query circuits by trigonometric polynomials. Section 3 separates finite known supports from interval promises. In Section 4, global unitarity forces a nonnegative Taylor limit; exterior Chebyshev growth computes its best approximation error; and a polynomial gate pinned at the extremal contacts turns that limiting solution into a globally contractive converter without exceeding the threshold error. Section 5 replaces fixed-order compactness by uniform Taylor estimates and an independent Fejér–Riesz argument, then controls the pinned construction for growing order. Section 6 gives the LP realization and separates state preparation from coherent compilation. Section 7 states the remaining questions, and Appendix A describes the reproducible figures and source build.

2 Access model and polynomial implementation

An exact block encoding of a Hermitian contraction H is a unitary U and a known isometry $J: v \mapsto |0^a\rangle \otimes v$, with $J^*UJ = H$. A unit-normalized η -approximate block encoding of A is a unitary V and a known isometry J_{out} such that

$$\|J_{\text{out}}^* V J_{\text{out}} - A\| \leq \eta.$$

More generally, an exact block encoding of A with normalization $\nu > 0$ satisfies $J_{\text{out}}^* V J_{\text{out}} = A/\nu$. All norms of operators are spectral norms. A converter is a circuit whose gates may depend on the stated spectral promise and accuracy, but not on the unknown matrix or on its oracle completion. It must work for every unitary U with the prescribed compressed block. One controlled call to U or U^* costs one query. Ancillas, inverse calls, and coherently controlled choices of queries are allowed. The output is a reusable unitary: there is no conditioning on a favorable measurement outcome. Intermediate measurements with a bounded worst-case query count can be deferred and their classical controls made coherent. The lower bounds below therefore include that form of adaptivity. We do not consider expected-cost algorithms with unbounded stopping times.

The principal promise is

$$0 < \delta, \quad \rho > 1, \quad 0 < c \leq 1, \quad \rho\delta < c, \quad \text{spec}(H) \subseteq [\delta, \rho\delta] \cup [c, 1]. \quad (1)$$

Let $Q(\delta, \eta; \rho, c)$ be the minimum worst-case query count for unit-normalized conversion to $I - H$ under this promise, uniformly in matrix dimension. Constants in asymptotic notation may depend on fixed parameters indicated by subscripts. In the fixed-relative-accuracy results, ρ, c, K are fixed and $\eta = K\delta$ as $\delta \downarrow 0$.

Lemma 2.1 (Bounded polynomial implementation). *If $p \in \mathbb{R}[x]$ has degree d and $|p(x)| \leq 1$ on $[-1, 1]$, then $p(H)$ has an exact unit-normalized block encoding using $O(d)$ controlled queries to any exact block encoding of the Hermitian contraction H . If p has definite parity, d queries suffice in the real-polynomial QSVT convention.*

Proof. The definite-parity statement is the real-polynomial singular value transformation theorem of Gilyén et al. [9, Corollary 18, extended version]. It coherently averages the phase sequences with opposite signs to extract the real part, using an extra qubit but no normalization loss. For

any Hermitian H , the even transform gives $p(|H|) = p(H)$, while the odd transform includes the left/right singular-vector signs and gives $\text{sign}(H)p(|H|) = p(H)$. On the zero eigenspace the odd case follows from $p(0) = 0$.

For completeness, the general-parity reduction does not require assumptions on the unknown completion. Define a Hermitian unitary

$$\tilde{U} = |0\rangle\langle 1| \otimes U + |1\rangle\langle 0| \otimes U^*, \quad \tilde{J} = |+\rangle \otimes J.$$

Its compression is H , and it costs a constant number of controlled oracle calls. Set $\Pi = \tilde{J}\tilde{J}^*$ and $V = (2\Pi - I)\tilde{U}$. For each eigenvector v of H with eigenvalue $x \in (-1, 1)$, the span of $\tilde{J}v$ and $(\tilde{U}\tilde{J}v - x\tilde{J}v)/\sqrt{1-x^2}$ is invariant, and V has matrix

$$\begin{pmatrix} x & \sqrt{1-x^2} \\ -\sqrt{1-x^2} & x \end{pmatrix}.$$

Thus $(V + V^*)/2 = xI$ on this subspace; the endpoint eigenspaces follow directly. The ordinary polynomial

$$R(z) = z^d p((z + z^{-1})/2)$$

has degree at most $2d$ and modulus at most one on the unit circle. Generalized quantum signal processing (GQSP) [15], via a Fejér–Riesz completion, implements $R(V)$ with at most $2d$ controlled walk queries. Multiplication by V^{-d} uses another d queries and gives $p((V + V^*)/2)$. Compression by $\tilde{J}$ yields $p(H)$. This construction is coherent and unit-normalized throughout. $\square$

Lemma 2.2 (Scalar representation and derivative bounds). *For the scalar oracle*

$$U(t) = \begin{pmatrix} \sin t & \cos t \\ \cos t & -\sin t \end{pmatrix}, \tag{2}$$

each matrix element $q(t)$ of a T -query converter is a trigonometric polynomial of degree at most T and satisfies $|q(t)| \leq 1$ for every real t . The function $g(x) = 1 - \text{Re } q(\arcsin x)$ obeys $0 \leq g \leq 2$ on $[-1, 1]$ and, for each fixed integer $j \geq 1$,

$$\sup_{|x| \leq 1/2} |g^{(j)}(x)| \leq C_j(T+1)^j. \tag{3}$$

Proof. Each query has entries of trigonometric degree at most one, including a controlled query, whose inactive block is the identity. Multiplication and addition of entries increase the degree by at most one per query. Oracle-independent gates and ancillas do not change this induction. Unitarity bounds each output entry for all t , whether or not $\sin t$ satisfies the correctness promise. Trigonometric Bernstein inequalities bound the j th derivative in t by $2T^j$ for $1 - \text{Re } q(t)$. Repeated differentiation of $t = \arcsin x$, whose derivatives of every fixed order are bounded on $[-1/2, 1/2]$, proves (3). $\square$

3 Exact supports and the normalization boundary

Proposition 3.1 (Two-query conversion). *Let $0 < a < b \leq 1$ be known, with $(1-a)(1-b) \leq a+b$. The polynomial*

$$p_{a,b}(x) = 1 - \frac{ab + x^2}{a + b}$$

is bounded in absolute value by one on $[-1, 1]$. Two QSVT queries implement it, and

$$p_{a,b}(H) - (I - H) = -\frac{(H - aI)(H - bI)}{a + b}. \quad (4)$$

Consequently the conversion is exact on $\text{spec}(H) \subseteq \{a, b\}$; on any other spectral support its error is exactly $\max_{x \in \text{spec}(H)} |(x - a)(x - b)|/(a + b)$.

Proof. The polynomial decreases as x^2 increases. Its maximum is $1 - ab/(a + b) < 1$, and its minimum is $-(1 - a)(1 - b)/(a + b) \geq -1$. Lemma 2.1 and factorization give the assertions. $\square$

In particular, $b \geq 1/2$ suffices for the hypothesis, and $b = 1$ gives $p_{a,1}(x) = (1 - x^2)/(1 + a)$. Thus a large condition number, by itself, does not force an expensive normalized shift. This result concerns conversion of an operator; it does not remove the normalization and overlap costs of a linear-system output state.

Proposition 3.2 (An interval cannot be converted exactly). *No finite-query converter produces an exact encoding of $I - H$ with normalization $0 < \nu < 2$ for every scalar $H = x$ in any nonempty open subinterval of $(0, 1)$.*

Proof. By Lemma 2.2, its designated output entry $q(t)$ is a trigonometric polynomial. Equality with $(1 - \sin t)/\nu$ on an open interval implies equality for every real t , by analyticity. At $t = -\pi/2$ this would give $q(t) = 2/\nu > 1$, contradicting unitarity. $\square$

Proposition 3.3 (Pairwise obstruction and normalization slack). *Suppose $0 < \delta \leq 1/4$ and a converter works at both $H = \delta$ and $H = 2\delta$ with error at most $\delta/16$. It uses at least*

$$\frac{\sqrt{3}}{2} \left(\sqrt{\frac{93}{32}} - \sqrt{\frac{17}{8}} \right) \delta^{-1/2}$$

queries. The formal differentiated hybrid bound says that if an exact converter with normalization ν works on a neighborhood of $x = \delta$, where $\nu > 1 - \delta$, then

$$T \geq \frac{(1 - \delta)\sqrt{1 - \delta^2}}{\nu\sqrt{\nu^2 - (1 - \delta)^2}}. \quad (5)$$

Proof. Use the canonical oracle $U(\arcsin x)$. Its derivative in x has norm $1/\sqrt{1 - x^2}$, so the hybrid argument gives

$$\|V_{2\delta} - V_\delta\| \leq \frac{2T\delta}{\sqrt{3}}.$$

Apply each output unitary to its designated input basis vector. If the designated output amplitude is z , the norm of the complementary component is $\sqrt{1 - |z|^2}$. At $x = \delta$ this norm is at most $\sqrt{17\delta/8}$, since $|z| \geq 1 - 17\delta/16$. At $x = 2\delta$ it is at least $\sqrt{93\delta/32}$: indeed $|z| \leq 1 - 31\delta/16$, and $1 - (1 - s)^2 \geq 3s/2$ for $s = 31\delta/16 \leq 1/2$. The difference between the complementary norms is a lower bound on the distance of the output vectors, proving the first claim.

For the second claim the exact designated amplitude is $f(x) = (1 - x)/\nu$. The complementary norm has derivative $(1 - x)/(\nu\sqrt{\nu^2 - (1 - x)^2})$. Compare neighboring inputs, apply the hybrid argument with oracle derivative $1/\sqrt{1 - x^2}$, divide by their separation, and take the limit. $\square$

The pairwise method is a specialization of the eigenvalue-transformation lower-bound method in [9, Theorem 73, extended version]. The differentiated inequality records the local geometric obstruction, but it does not describe an attainable exact tradeoff near normalization one: Proposition 3.2 already rules out exact conversion for every $\nu < 2$. At normalization two, a direct coherent linear combination implements $(I - H)/2$ with constant queries. The accuracy-dependent cost at normalization one is the substantive constraint studied below.

4 The complete fixed-relative-accuracy staircase

All approximation-order indices, including r and j , are nonnegative integers. For $r \geq 0$ define

$$G_r(\rho) = \inf \left\{ \max_{1 \leq y \leq \rho} |P(y) - y| : P \in \mathbb{R}[y], \deg P \leq 2r, P \geq 0 \text{ on } \mathbb{R} \right\}. \quad (6)$$

The constraint on the entire real line is essential: the approximation error is measured only on $[1, \rho]$.

Theorem 4.1 (Optimal fixed-accuracy queries). *Fix $\rho > 1$, $0 < c \leq 1$, and $K > 0$, and put $\ell = \min\{j \geq 0 : G_j(\rho) \leq K\}$. Then ℓ is finite and, as $\delta \downarrow 0$,*

$$Q(\delta, K\delta; \rho, c) = \begin{cases} \Theta_{\rho, c, K}(\log(1/\delta)), & \ell = 0, \quad c < 1, \\ \Theta_{\rho, K}(1), & \ell = 0, \quad c = 1, \\ \Theta_{\ell, \rho, c, K}(\delta^{-1+1/(2\ell)}), & \ell \geq 1. \end{cases} \quad (7)$$

Every threshold equality $K = G_\ell$ belongs to the cheaper tier displayed in this formula.

The distinction at $c = 1$ has a simple cause: $[c, 1]$ is then a single known spectral point. A high interval of positive length supplies an additional logarithmic obstruction, whereas one known point can be handled by interpolation.

4.1 The globally nonnegative approximation problem

Write

$$m_\rho = \frac{\rho + 1}{2}, \quad h_\rho = \frac{\rho - 1}{2}, \quad z_0 = \frac{m_\rho}{h_\rho},$$

and let T_n denote the Chebyshev polynomial, $T_n(\cos \theta) = \cos(n\theta)$.

Theorem 4.2 (Exact thresholds). *We have $G_0 = h_\rho$. For $r \geq 1$,*

$$G_r = h_\rho \max_{z \geq z_0} \frac{z - z_0}{T_{2r}(z)}, \quad (8)$$

$$P_r^*(y) = y + G_r T_{2r}((y - m_\rho)/h_\rho) \quad (9)$$

are respectively the optimal error and an admissible minimizer. The positive numbers G_r strictly decrease to zero. If z_ is the maximizer in (8), it is unique and*

$$z_* - z_0 = T_{2r}(z_*)/T'_{2r}(z_*), \quad G_r = h_\rho/T'_{2r}(z_*). \quad (10)$$

Proof. The constant case is ordinary interval minimax approximation. For $n = 2r \geq 2$, the exterior Chebyshev inequality says that a degree-at-most- n polynomial A bounded by one on $[-1, 1]$ satisfies $|A(z)| \leq T_n(|z|)$ for $|z| > 1$. One elementary proof compares A with T_n at its $n + 1$ successive extrema: if the exterior inequality failed, subtracting a suitable multiple of T_n would give n interior sign changes and an exterior zero, contradicting the degree. Apply this inequality to $P(y) - y$. At $y = -h_\rho(z - z_0) \leq 0$, nonnegativity gives $P(y) - y \geq h_\rho(z - z_0)$, proving the lower bound in (8).

To prove admissibility of (9), set $e = G_r/h_\rho$ and $x = (y - m_\rho)/h_\rho$. We must show $x + z_0 + eT_n(x) \geq 0$ for every real x . For $x \leq -z_0$ this is precisely the maximizing inequality defining e . For $-z_0 \leq x \leq -1$ and for $x \geq 1$, both terms are nonnegative. On $[-1, 1]$, only negative lobes of T_n require attention. They satisfy $x \geq -\cos(\pi/(2n))$, so

$$x + z_0 > 1 - \cos(\pi/(2n)) \geq n^{-2}.$$

The last bound follows from $\sin u \geq 2\sqrt{2}u/\pi$ on $[0, \pi/4]$. Since T_n is strictly convex on $[1, \infty)$ and $T'_n(1) = n^2$, $T_n(z) \geq 1 + n^2(z - 1)$, whence $e < n^{-2}$. Since $T_n(x) \geq -1$, positivity follows. The approximation error is exactly G_r by Chebyshev alternation.

The maximum in (8) is positive, attained away from z_0 and infinity. The function $z - T_n(z)/T'_n(z)$ increases strictly on $(1, \infty)$, with derivative $T_n T''_n / (T'_n)^2 > 0$, and tends to infinity. This proves uniqueness and (10). For successive positive indices, evaluate the ratio for T_{2r+2} at its maximizer and use $T_{2r+2}(z) > T_{2r}(z)$ for $z > 1$. Also $G_1 < G_0$: the explicit formula below proves this directly. Finally, polynomial approximations to $\sqrt{y}$ on $[1, \rho]$, squared, are globally nonnegative and converge uniformly to y . Thus $G_r \rightarrow 0$. $\square$

The admissibility has a geometric antecedent: by (10), in the coordinate $z = (m_\rho - y)/h_\rho$, the extremizer is G_r times the difference between $T_{2r}(z)$ and its tangent at z_* . Its nonnegativity also follows from the supporting-tangent inequality of Fawzi, Saunderson, and Parrilo [8, Appendix A, Lemma 1]; the proof above is self-contained.

For the first threshold the formula simplifies to

$$G_1 = \frac{\rho + 1 - \sqrt{(\rho^2 + 6\rho + 1)/2}}{4}, \quad P_1^*(y) = \frac{(y + s)^2}{2(m_\rho + s)}, \quad s = \sqrt{m_\rho^2 - h_\rho^2/2}. \quad (11)$$

These expressions follow either by solving (10) for $T_2(z) = 2z^2 - 1$ or by expanding (9).

Proposition 4.3 (Threshold asymptotics). *Let $R_0 = (\sqrt{\rho} + 1)/(\sqrt{\rho} - 1)$ and $a = \log R_0$. As $r \rightarrow \infty$,*

$$G_r = \frac{\sqrt{\rho}}{e^r} R_0^{-2r} (1 + O_\rho(r^{-1})). \quad (12)$$

Consequently, as $K \downarrow 0$, with $L = \log(1/K)$,

$$\ell(K) = \frac{L - \log L + \log(2\sqrt{\rho} a/e)}{2a} + O_\rho(1), \quad 1 - \frac{1}{2\ell(K)} = 1 - \frac{a}{L - \log L + O_\rho(1)}.$$

The $O(1)$ term in the first expression includes integer rounding.

Proof. Put $n = 2r$, $z = \cosh t$, and note $z_0 = \cosh a$. The stationary equation is

$$n(\cosh t - \cosh a) \tanh(nt) = \sinh t.$$

Its solution has $t = a + O_\rho(n^{-1})$: indeed the increasing left-to-right ratio $n(\cosh t - \cosh a)/\sinh t$ crosses one within such a neighborhood, and $\tanh(nt) = 1 + O_\rho(e^{-2na})$. Expanding with $t = a + s/n$ now gives $s = 1 + O_\rho(n^{-1})$. Substitution into (8), using $h_\rho \sinh a = \sqrt{\rho}$, gives (12). Taking logarithms and inverting $L = 2ar + \log r - \log(\sqrt{\rho}/e) + O_\rho(1/r)$ gives the last claims, with bounded rounding uncertainty. $\square$

4.2 An all-circuit lower bound

Theorem 4.4 (Nonnegative Taylor limit). *Fix $r \geq 0$ and $K < G_r(\rho)$. Any coherent converter accurate to $K\delta$ on $[\delta, \rho\delta]$ uses*

$$T = \Omega_{r,\rho,K}(\delta^{-(2r+1)/(2r+2)})$$

queries. No upper spectral interval is needed for this bound.

Proof. Use Lemma 2.2 and put $g_\delta(x) = 1 - \operatorname{Re} q_\delta(\arcsin x)$. Correctness implies

$$|g_\delta(\delta y)/\delta - y| \leq K \quad (1 \leq y \leq \rho).$$

Taylor expansion through order $2r + 1$, followed by rescaling, gives a polynomial A_δ of degree at most $2r + 1$ such that, on each fixed compact set of real y ,

$$g_\delta(\delta y)/\delta = A_\delta(y) + O_{r,y}((T + 1)^{2r+2}\delta^{2r+1}).$$

If the claimed bound failed, there would be a sequence $\delta \downarrow 0$ with $T\delta^{(2r+1)/(2r+2)} \rightarrow 0$, so the remainder would vanish uniformly on each fixed compact set. Values at $2r + 2$ fixed distinct points of $(1, \rho)$ then bound all coefficients of A_δ by invertibility of the Vandermonde matrix. Pass to a coefficientwise convergent subsequence, with limit A . For each fixed real y , global nonnegativity of g_δ gives $A(y) \geq 0$. A nonzero nonnegative polynomial has even degree; thus $\deg A \leq 2r$. It approximates y within K on $[1, \rho]$, contradicting the definition of G_r . $\square$

If $\eta(\delta)/\delta \rightarrow 0$, this immediately yields $T = \Omega_{\rho,\varepsilon}(\delta^{-1+\varepsilon})$ for every fixed $\varepsilon > 0$ (with the onset depending on the accuracy function). This is stronger than the pairwise square-root obstruction. It uses contractivity outside the correctness interval, through the globally nonnegative Taylor limit.

4.3 Bounded constructions, including exact threshold errors

We first record a positive approximant for the fixed upper interval. Define d_j by

$$\frac{1 - \sqrt{1 - z}}{1 - z} = \sum_{j \geq 1} d_j z^j, \quad S_N(u) = u \sum_{j=1}^N d_j (1 - u)^j.$$

The coefficients d_j are partial sums of the positive Taylor coefficients of $1 - \sqrt{1 - z}$, so $0 < d_j \leq 1$. For $0 \leq u \leq 1$,

$$0 \leq S_N(u) \leq 1 - \sqrt{u}, \quad 0 \leq 1 - \sqrt{u} - S_N(u) \leq (1 - u)^{N+1}, \quad S_N(u) \leq Nu. \quad (13)$$

The tail estimate follows by comparison with the geometric series; the endpoints follow by continuity. For $c < 1$, $N = O_c(\log(1/\delta))$ makes the high-band tail any prescribed constant multiple of δ . For $c = 1$, $N = 1$ suffices because $S_N(1) = 0$.

Lemma 4.5 (A pinned gate). *Fix $r \geq 1$ and a nonempty finite set $A \subseteq [1, \rho]$. Put $\alpha = 1 - 1/(2r)$ and choose an odd integer $m \asymp M\delta^{-\alpha}$, where $M > 0$ is fixed. There is an even polynomial W of degree $O_{r,|A|}(m)$ with $0 \leq W \leq 1$ on $[-1, 1]$, $W(\delta y) = 1$ for $y \in A$, and*

$$1 - W(\delta y) \leq C(m\delta)^{2r+2} \operatorname{dist}(y, A)^{2r+2}, \quad 1 \leq y \leq \rho, \quad (14)$$

$$W(x) \leq C(m|x|)^{-2r-4}, \quad C_0\delta \leq |x| \leq 1. \quad (15)$$

Constants may depend on r, ρ, A , but not on δ, m ; C_0 is chosen sufficiently large depending on ρ .

Proof. For $\theta_j = \arcsin(\delta y_j)$, $y_j \in A$, define

$$D_{m,a}(\theta) = \frac{\sin(m(\theta - a))}{m \sin(\theta - a)}, \quad W(\sin \theta) = 1 - \prod_{y_j \in A} \prod_{\sigma = \pm 1} \left(1 - D_{m,\sigma\theta_j}(\theta)^{2r+4}\right)^{r+1}.$$

Removable singularities are filled by continuity. The elementary inequality $|\sin(mu)| \leq m|\sin u|$ gives $|D_{m,a}| \leq 1$. For odd m , $D_{m,a}$ is a real trigonometric polynomial with even frequencies and

degree $m - 1$. Pairing a with $-a$ makes the product even in θ and π -periodic. It is therefore a polynomial in $\cos 2\theta = 1 - 2x^2$, of degree $O_{r,|A|}(m)$ in x . The range and pinning properties are immediate.

Near a pin, $1 - D_{m,a}(\theta)^{2r+4} \leq C_r m^2 (\theta - a)^2$. Select the nearest positive pin in the product defining $1 - W$ and bound all other factors by one. Since $|\arcsin(\delta y) - \arcsin(\delta y_j)| \leq 2\delta|y - y_j|$ for small δ , this proves (14). Finally, if $C_0\delta \leq |x| \leq 1$, then $|\sin(\arcsin x \pm \theta_j)| \geq |x|/2$. Hence $|D_{m,\pm\theta_j}| \leq 2/(m|x|)$. Applying $1 - \prod_i (1 - a_i)^{r+1} \leq (r+1) \sum_i a_i$ proves (15). $\square$

Proposition 4.6 (Upper bound at every positive threshold). *For each fixed $r \geq 1$ there is a polynomial q_δ bounded by one on $[-1, 1]$ with error at most $G_r\delta$ on (1) and degree $O_{r,\rho,c}(\delta^{-1+1/(2r)})$.*

Proof. Take $P = P_r^*$ and $e(y) = P(y) - y$. The positive-contact set $A = \{y \in [1, \rho] : e(y) = G_r\}$ is finite and nonempty, by (9). Use Lemma 4.5 for this set and define

$$q_\delta(x) = W(x)[1 - \delta P(x/\delta)] + (1 - W(x))S_N(x^2). \quad (16)$$

We verify contractivity before its approximation property. Write $q_\delta = B - \delta P(x/\delta)W$ with $B = W + (1 - W)S_N(x^2) \in [0, 1]$. Thus $q_\delta \leq 1$. Since $P(y) \leq C(1 + |y|^{2r})$ and $P \geq 0$, splitting at $|x| = C_0\delta$ and $|x| = 1/m$ gives

$$0 \leq \delta P(x/\delta)W(x) \leq C\delta + C\delta^{1-2r}m^{-2r} \leq C\delta + C'M^{-2r}. \quad (17)$$

Indeed $C_0\delta < 1/m$ for small δ ; on the middle region use $W \leq 1$, and on the outer region use (15). Choose M large enough and then δ small enough to make (17) at most one. It follows that $q_\delta \geq -1$.

For the lower interval, the polynomial $G_r - e(y)$ is nonnegative on $[1, \rho]$ and has zeros there exactly at A , with multiplicities at most $2r$. Compactness and factorization therefore give a constant $b > 0$ such that

$$G_r - e(y) \geq b \operatorname{dist}(y, A)^{2r} \quad (1 \leq y \leq \rho).$$

Since $(m\delta)^{2r+2} = O(\delta^{1+1/r})$, (14) and $W = 1 - o(1)$ on this interval imply, for small δ ,

$$1 - W(\delta y) \leq \frac{1}{2}W(\delta y)\delta[G_r - e(y)]. \quad (18)$$

This estimate includes the contacts themselves, where both sides vanish. Put $d(x) = 1 - x - S_N(x^2) \in [0, 1]$ for $0 \leq x \leq 1$. On $x = \delta y$,

$$(1 - x) - q_\delta(x) = W\delta e(y) + (1 - W)d(x).$$

Its lower bound is $-G_r\delta$ because $e \geq -G_r$ and $d \geq 0$. Its upper bound follows from (18):

$$G_r\delta - [(1 - x) - q_\delta(x)] = W\delta(G_r - e) + (1 - W)(G_r\delta - d) \geq 0.$$

Thus the error is at most exactly $G_r\delta$, without an asymptotic error overshoot.

On $[c, 1]$, the tail in (13) can be made at most $G_r\delta/3$. Also $W = O(m^{-2r-4}) = o(\delta)$, and

$$\delta P(x/\delta)W = O(\delta^{1-2r}m^{-2r-4}) = O(\delta^{4-2/r}) = o(\delta).$$

These two terms can each be made at most $G_r\delta/3$ by decreasing δ . Finally, (16) has degree $O_{r,\rho}(m + N)$, proving the claim. $\square$

The pinned construction handles strict bands as well, by using the smallest r with $G_r \leq K$. Pinning is necessary in this proof to control the sign of the error at the exact minimax contacts; a gate with merely $o(\delta)$ leakage would only establish errors $(G_r + o(1))\delta$.

4.4 The coarse tier and completion of the staircase proof

Proposition 4.7 (Coarse conversion). *For $K \geq G_0 = (\rho - 1)/2$, the first two lines of (7) hold.*

Proof. It suffices to construct error $G_0\delta$. Set $P = m_\rho$, so $e(y) = m_\rho - y$ and $G_0 - e(y) = y - 1$. First suppose $c < 1$ and use

$$W(x) = [1 - (x^2 - \delta^2)^2]^L, \quad L = \lceil C_c \log(1/\delta) \rceil.$$

For small δ , the base is in $[0, 1]$ on $[-1, 1]$. On the lower band,

$$1 - W(\delta y) \leq L\delta^4(y^2 - 1)^2 \leq C_\rho L\delta^4(y - 1)^2.$$

This is at most $\frac{1}{2}W\delta(y - 1)$ for sufficiently small δ . On the high band, $W \leq (1 - c^4/4)^L$ for δ sufficiently small. Choose the constant in L so that this leakage is at most $G_0\delta/3$, and choose $N = O_c(\log(1/\delta))$ so the tail in (13) is at most $G_0\delta/3$. The blend (16) is a convex combination of $1 - m_\rho\delta$ and $S_N(x^2)$, hence lies in $[0, 1]$. The same signed-error argument as in Proposition 4.6 proves the exact low-band error. On the high band its error is at most the tail plus W , which is below $G_0\delta$. Its degree is $O(\log(1/\delta))$.

If $c = 1$, instead take

$$W(x) = 1 - \left(\frac{x^2 - \delta^2}{1 - \delta^2} \right)^2, \quad q_\delta(x) = (1 - m_\rho\delta)W(x).$$

For small δ , $0 \leq W \leq 1$ on $[-1, 1]$, $W(\delta) = 1$, and $W(1) = 0$. The low leakage is $O_\rho(\delta^4(y - 1)^2)$ and the same signed-error argument applies with $S_N = 0$. This degree-four polynomial is exact at the high point 1. A zero-query circuit has the same scalar output at $x = \delta$ and $x = 1$, whose targets differ by $1 - \delta$, so cannot satisfy the error contract for small δ . This gives $\Theta(1)$.

For the logarithmic lower bound when $c < 1$, choose a closed interval I of positive length b strictly inside $(\arcsin c, \pi/2)$. Let $q(t)$ be the scalar output entry and $R(t) = q(t) - (1 - \sin t)$. It has trigonometric degree $n \leq \max(T, 1)$, is bounded by $K\delta$ on I , and satisfies $|R(-\pi/2)| \geq 1$ by contractivity. Here is a direct form of the needed interpolation inequality. If R has degree at most n , then $z^n R(t)$, $z = e^{it}$, is an ordinary polynomial of degree at most $2n$. Interpolate at $2n + 1$ equally spaced angles in I . For node indices i, j , their distance on the unit circle is at least $b|i - j|/(\pi n)$, whereas its distance to any evaluation point on the unit circle is at most two. The sum of the absolute Lagrange weights is consequently at most

$$\left(\frac{2\pi n}{b} \right)^{2n} \sum_{j=0}^{2n} \frac{1}{j!(2n - j)!} = \frac{(4\pi n/b)^{2n}}{(2n)!} \leq (2\pi e/b)^{2n}.$$

Thus $1 \leq K\delta(2\pi e/b)^{2n}$, giving $T = \Omega_{c,K}(\log(1/\delta))$. $\square$

Proof of Theorem 4.1. Theorem 4.2 guarantees existence of ℓ . For $\ell \geq 1$, the lower bound is Theorem 4.4 with $r = \ell - 1$, since $K < G_{\ell-1}$. Proposition 4.6, followed by Lemma 2.1, gives the matching upper bound because $G_\ell \leq K$. Proposition 4.7 treats $\ell = 0$. $\square$

4.5 Definite parity and a polynomial query separation

Even polynomials satisfy additional lower bounds governed by the following approximation errors. Define

$$E_r(\rho) = \inf_{A \in \mathbb{R}[v], \deg A \leq r} \max_{1 \leq y \leq \rho} |y - A(y^2)|.$$

These unconstrained errors first give elementary lower-bound cutoffs. Retaining global nonnegativity below will give an optimal implicit even-parity staircase, including equality cases. Each E_r is positive: otherwise finite-dimensional compactness would give an exact polynomial representation of $\sqrt{v}$ on $[1, \rho^2]$, which is impossible. In particular, the affine cutoff is

$$E_1 = \frac{(\rho - 1)^2}{8(\rho + 1)}, \quad A_1(v) = a_\rho v + b_\rho, \quad a_\rho = \frac{1}{\rho + 1}, \quad b_\rho = \frac{\rho}{\rho + 1} + E_1. \quad (19)$$

To verify minimax optimality, the error $\sqrt{v} - A_1(v)$ equals $-E_1$ at $v = 1, \rho^2$, and E_1 at $v = (\rho + 1)^2/4$. An affine perturbation reducing all three absolute errors would need two sign changes, which is impossible.

Proposition 4.8 (Parity obstructions). *Let p be a real polynomial bounded by one on $[-1, 1]$ that approximates $1 - x$ within $K\delta$ on $[\delta, \rho\delta]$. If p is even and $K < E_r(\rho)$, then*

$$\deg p = \Omega_{r,\rho,K}(\delta^{-1+1/(2r+2)}).$$

More explicitly, if $\deg p = 2m$ and $K < E_1$, then

$$m \geq \left(\frac{3(E_1 - K)}{2\rho^4} \right)^{1/4} \delta^{-3/4}. \quad (20)$$

If p is odd, then

$$\deg p \geq \frac{[1 - (1 + K)\delta]\sqrt{1 - \delta^2}}{\delta}.$$

Proof. Taylor-expand an even degree- d polynomial at zero through degree $2r$. On a fixed interior subinterval, Bernstein inequalities give $|p^{(j)}| \leq C_j(d+1)^j$ for each fixed j . The vanishing odd Taylor terms imply that $(1 - p(\delta y))/\delta$ differs from a degree- r polynomial in y^2 by at most $C_{r,\rho}(d+1)^{2r+2}\delta^{2r+1}$. Therefore $E_r \leq K + C_{r,\rho}(d+1)^{2r+2}\delta^{2r+1}$, giving the first claim. For the explicit constant, write $p(x) = A(x^2)$ with $\deg A = m$ and $\|A\|_{[0,1]} \leq 1$. The second Markov inequality on $[0, 1]$ gives $\|A''\| \leq 4m^2(m^2 - 1)/3$. The first-order Taylor remainder at $u = \delta^2 v$, divided by δ , is at most $2m^4\rho^4\delta^3/3$. Comparing to the best affine error proves (20). Finally, odd parity gives $p(0) = 0$, so the mean-value theorem and Bernstein's first-derivative inequality give $(1 - (1 + K)\delta)/\delta \leq \sup_{[0,\delta]} |p'| \leq d/\sqrt{1 - \delta^2}$. $\square$

Proposition 4.9 (Even square-root construction). *For every fixed $\gamma > 0$, there is an even polynomial bounded by one on $[-1, 1]$ with error at most $(E_1 + \gamma)\delta$ on (1) and degree $O_{\rho,c,\gamma}(\delta^{-1/2})$.*

Proof. For an integer $m \geq 2$ define the normalized endpoint Fejér kernel

$$F_m(u) = \frac{1 - T_m(1 - 2u)}{2m^2u}, \quad F_m(0) = 1.$$

This is a polynomial of degree $m - 1$. If $u = \sin^2(\theta/2)$, then $F_m(u) = \sin^2(m\theta/2)/(m^2 \sin^2(\theta/2))$, so $0 \leq F_m \leq 1$ on $[0, 1]$ and $F_m(u) \leq 1/(m^2u)$ for $0 < u \leq 1$. Taylor expansion at zero gives

$$F_m(u) = 1 - \frac{m^2 - 1}{3}u + O(m^4u^2).$$

The remainder is uniform when $m^2u = o(1)$; it also follows from the second Markov bound and $\|F_m\|_{[0,1]} \leq 1$. Choose an integer m nearest the positive solution of $2(m^2 - 1)\delta/3 = a_\rho$ and set $W_m = F_m^2$. Integer rounding gives, uniformly on $1 \leq v \leq \rho^2$,

$$1 - W_m(\delta^2 v) = a_\rho \delta v + O_\rho(\delta^{3/2}).$$

Using (13) with a high-band tail $O(\delta^2)$, define

$$P_\delta(u) = (1 - b_\rho \delta)W_m(u) + (1 - W_m(u))S_N(u), \quad p_\delta(x) = P_\delta(x^2).$$

This is a convex combination of numbers in $[0, 1]$. On the low band, $S_N(\delta^2 v) = O(N\delta^2)$ and hence

$$1 - P_\delta(\delta^2 v) = \delta(b_\rho + a_\rho v) + O_{\rho,c}(\delta^{3/2} + \delta^2 \log(1/\delta)).$$

On the high band, $W_m(u) \leq 1/(m^4 u^2) = O_{\rho,c}(\delta^2)$, so $P_\delta(u) = 1 - \sqrt{u} + O_{\rho,c}(\delta^2)$. The degree is at most $2[2(m-1) + N + 1] = O_{\rho,c}(\delta^{-1/2})$, and (19) finishes the proof. $\square$

Definition 4.10 (Nonnegative even thresholds). For $r \geq 0$ let

$$F_r(\rho) = \inf_{\substack{P \in \mathbb{R}[y] \text{ even, } \deg P \leq 2r \\ P \geq 0 \text{ on } \mathbb{R}}} \max_{1 \leq y \leq \rho} |P(y) - y|.$$

These constrained thresholds differ from the unconstrained errors E_r . The distinction strengthens both the general parity hierarchy and the concrete comparison below.

Theorem 4.11 (Optimal even-parity staircase). *The numbers $F_r > 0$ are attained, nonincreasing, and tend to zero. For fixed ρ, c, K , let $\ell = \min\{r : F_r \leq K\}$. The minimum degree of a real even polynomial bounded by one on $[-1, 1]$ and approximating $1 - x$ within $K\delta$ on (1) has the orders in (7), with this value of ℓ . In particular,*

$$F_0 = G_0, \quad F_1 = F_2 = E_1. \tag{21}$$

Thus $K = E_1$ admits degree $\Theta(\delta^{-1/2})$, whereas $K < E_1$ forces degree $\Omega(\delta^{-5/6})$.

Proof. A minimizing sequence has bounded values at $2r + 1$ fixed distinct points of $[1, \rho]$, hence bounded coefficients. Its coefficientwise limit is even and nonnegative on $\mathbb{R}$, proving attainment. Zero error would identify an even polynomial with y on an interval, an impossibility. Squaring polynomial approximations to $v^{1/4}$ on $[1, \rho^2]$, and then substituting $v = y^2$, proves convergence to zero. Inclusion of the feasible sets proves monotonicity; plateaus are allowed.

For the lower bound with $K < F_r$, repeat the compactness proof of Theorem 4.4 for the rescaled defect $(1 - p(\delta y))/\delta$. For each fixed derivative order j , interior polynomial Bernstein inequalities give $|p^{(j)}| \leq C_j(d+1)^j$, where $d = \deg p$. Taylor expansion through order $2r + 1$ therefore has remainder $O((d+1)^{2r+2}\delta^{2r+1})$ after rescaling, and all its odd coefficients vanish. If $d = o(\delta^{-1+1/(2r+2)})$, a coefficientwise subsequential limit is even, globally nonnegative, and of degree at most $2r$, contradicting $K < F_r$.

For the upper bound let P attain F_r , with $r \geq 1$. Its positive-error contact set $A = \{y : P(y) - y = F_r\}$ is nonempty: otherwise adding a sufficiently small positive constant would reduce the error while preserving nonnegativity. It is finite because $P(y) - y$ is nonconstant. The polynomial $F_r - P(y) + y$ is nonnegative on $[1, \rho]$; its zeros are exactly A , with multiplicities at most $2r$. Thus it is bounded below by $b \text{dist}(y, A)^{2r}$ on that interval for some $b > 0$. Lemma 4.5 and the proof of Proposition 4.6 now apply with this P and F_r in place of P_r^* and G_r : global growth is at most $C(1 + |y|^{2r})$, global nonnegativity proves contractivity, and pinning gives exact threshold error. The resulting blend is even because P , W , and $S_N(x^2)$ are even. Its degree is $O(\delta^{-1+1/(2r)})$. The coarse-tier constructions in Proposition 4.7 are also even, and its lower bounds apply to these transforms through Lemma 2.1. Minimality of ℓ gives $K < F_{\ell-1}$ and completes the staircase, including equality cases.

The positive affine polynomial $A_1(v)$ from (19) is nonnegative for $v \geq 0$, so $F_1 = E_1$. To prove $F_2 \geq E_1$, write any admissible even quartic as $P(y) = A(y^2)$ with $\deg A \leq 2$. Nonnegativity for $v \geq 0$ forces its quadratic coefficient to be nonnegative, so A is convex. Let $\ell(v)$ be the chord of $\sqrt{v}$ on $[1, \rho^2]$, and let the approximation error of A there be E . Convexity and the endpoint upper bounds give $A(v) \leq \ell(v) + E$. At $v_* = ((\rho + 1)/2)^2$, the lower error bound gives $A(v_*) \geq \sqrt{v_*} - E$. Since $\sqrt{v_*} - \ell(v_*) = 2E_1$, we have $E \geq E_1$. The admissible affine polynomial already attains equality, proving (21). $\square$

Proposition 4.12 (Optimal odd-parity order). *For fixed $\rho > 1$, $0 < c \leq 1$, and $K > 0$, the minimum degree of a real odd polynomial bounded by one on $[-1, 1]$ and accurate to $K\delta$ on (1) is*

$$\Theta_{\rho, K}(\delta^{-1} \log(1/\delta)).$$

The lower bound needs only the low interval, and the upper bound holds on all of $[\delta, 1]$.

Proof. Let p be such a polynomial of degree d . The trigonometric polynomial $q(t) = p(\sin t)$ has degree at most d , is bounded by one, and satisfies $q(0) = 0$. Set $t_1 = \arcsin \delta$, $t_2 = \arcsin(\rho\delta)$ and let A_n be its Taylor polynomial at zero through degree n . Bernstein's inequality and Taylor's theorem give

$$u := \|q - A_n\|_{[0, t_2]} \leq \frac{(dt_2)^{n+1}}{(n+1)!}.$$

On $[t_1, t_2]$, $|q(t) - 1| \leq (\rho + K)\delta$. The polynomial $1 - A_n$ has value one at zero, so exterior Chebyshev gives

$$1 \leq T_n\left(\frac{t_1 + t_2}{t_2 - t_1}\right) [(\rho + K)\delta + u].$$

The exterior coordinate tends to z_0 . Fix any $\bar{R} > R_0$; for all sufficiently small δ , the Chebyshev factor is at most $\bar{R}^n$, uniformly in n . Take $n = \lfloor \log(1/\delta)/(2 \log \bar{R}) \rfloor$. Then $(\rho + K)\delta \bar{R}^n \rightarrow 0$, and $u \geq \frac{1}{2} \bar{R}^{-n}$. Taking $(n+1)$ st roots and using the factorial bound yields $dt_2 \geq c_\rho n$. Since $t_2 = O_\rho(\delta)$, this proves the lower bound.

For the upper bound put $\tau = \min(\delta/2, K\delta/4)$ and $\epsilon = K\delta/4$. For small δ , the bounded odd sign approximant of [9, Lemma 25, extended version] gives s of degree $O(\tau^{-1} \log(1/\epsilon))$, with $|s| \leq 1$ on $[-1, 1]$ and $|s(x) - 1| \leq \epsilon$ for $x \geq \tau$. Define

$$p(x) = \frac{s(x) - x}{1 + \tau}.$$

For $0 \leq x \leq \tau$, its numerator is at least $-1 - \tau$; for $\tau \leq x \leq 1$, it is at least $-\epsilon$. Its numerator is at most one on $[0, 1]$. Thus $|p| \leq 1$ there for small δ , and oddness proves the same bound on $[-1, 1]$. For $x \geq \delta$,

$$|p(x) - (1 - x)| \leq \frac{\epsilon + \tau(1 - x)}{1 + \tau} \leq K\delta.$$

For fixed K , its degree has the stated order. $\square$

For $\rho = 2$ the elementary error $\delta/16$ lies above $F_1 = 1/24$, so Theorem 4.11 gives $\Theta_c(\delta^{-1/2})$ queries even within the even class. At the smaller error $\delta/32$,

$$G_1(2) = \frac{3 - \sqrt{17/2}}{4} \approx 0.021131 < \frac{1}{32} < \frac{1}{24} = F_1(2) = F_2(2).$$

To match the even lower bound, set

$$z = \frac{2}{5}y^2 - 1, \quad P(y) = \sqrt{\frac{5}{2}} \left(1 + \frac{z}{2} - \frac{z^2}{8} + \frac{z^3}{16} \right).$$

For real $y, z \geq -1$. The cubic in parentheses has derivative $1/2 - z/4 + 3z^2/16 > 0$ and value $5/16$ at $z = -1$, so P is globally positive. For $1 \leq y \leq 2, |z| \leq 3/5$. The binomial coefficients of $\sqrt{1+z}$ beyond degree three have absolute values at most $5/128$, and hence

$$|P(y) - y| \leq \sqrt{\frac{5}{2}} \frac{5}{128} \frac{(3/5)^4}{1 - 3/5} < \frac{1}{32}.$$

Thus $F_3(2) < 1/32$ and the even staircase has $\ell = 3$. We obtain the following fully matched comparison.

Converter class	Query order at $\rho = 2, \eta = \delta/32$
Arbitrary coherent converter (achieved by GQSP)	$\Theta(\delta^{-1/2})$
One even definite-parity QSVT transform	$\Theta(\delta^{-5/6})$
One odd definite-parity QSVT transform	$\Theta(\delta^{-1} \log(1/\delta))$

The comparison is with a single definite-parity transform, including the usual coherent extraction of its real part. It is not a lower bound on arbitrary compositions of QSVT circuits: such compositions are already allowed in the first row. All these statements count oracle queries and presume exact synthesis of oracle-independent gates; they do not claim a bound on classical phase-computation time or fault-tolerant gate synthesis.

5 Joint gap and accuracy limits

We now allow K to shrink with δ . Bounds whose constants depend on a fixed approximation order cannot simply be used with a growing order. We give two independent uniform lower bounds, a matching law at polynomial and higher accuracy, and a uniform pinned construction in the remaining regime. Throughout this section $\rho > 1$ is fixed,

$$D = \log(1/\delta), \quad L = \log(1/K), \quad R_0 = \frac{\sqrt{\rho} + 1}{\sqrt{\rho} - 1}, \quad a = \log R_0.$$

Constants denoted by C_ρ, c_ρ can change between occurrences, but never depend on a growing degree. A fixed constant raised to that degree is always displayed inside the power.

5.1 Exterior growth with the exact sine target

For every integer $n \geq 2$ set

$$H_n = h_\rho \max_{z \geq z_0} \frac{z - z_0}{T_n(z)}. \quad (22)$$

Thus $H_{2r} = G_r$, but odd indices also have a role. The proof of Proposition 4.3 applies without a parity assumption and gives

$$H_n = \frac{2\sqrt{\rho}}{en} R_0^{-n} (1 + O_\rho(n^{-1})). \quad (23)$$

In detail, the unique maximizing point is $z_n = \cosh(a + s_n/n)$ with $s_n = 1 + O_\rho(n^{-1})$, by the stationary equation $n(\cosh t - \cosh a) \tanh(nt) = \sinh t$. Consequently $a_n := h_\rho(z_n - z_0) = \Theta_\rho(1/n)$. Also H_n strictly decreases to zero: evaluate the ratio for H_{n+1} at its maximizer and use $T_{n+1}(z) > T_n(z)$ for $z > 1$.

Theorem 5.1 (Quantitative exterior lower bound). *There are $\delta_0, K_0 > 0$, depending only on ρ , such that the following holds for $0 < \delta < \delta_0$ and $0 < K < K_0$. Let n be the largest odd integer at least three with $H_n \geq 8K$. Every coherent converter accurate to $K\delta$ on $[\delta, \rho\delta]$ uses*

$$T \geq c_\rho n K^{1/(n+1)} \delta^{-n/(n+1)}. \quad (24)$$

Moreover,

$$n = \frac{L - \log L + O_\rho(1)}{a}, \quad K^{1/(n+1)} = \Theta_\rho(1). \quad (25)$$

The constants and the allowed δ_0 are uniform even when L grows arbitrarily fast.

Proof. On the scalar family of Lemma 2.2, put $F(t) = 1 - \operatorname{Re} q(t)$. Then F has trigonometric degree at most T , satisfies $0 \leq F \leq 2$ globally, and

$$|F(\arcsin x) - x| \leq K\delta \quad (\delta \leq x \leq \rho\delta).$$

Define

$$b_\delta = \frac{\arcsin(\rho\delta) - \arcsin \delta}{\rho - 1}, \quad a_\delta = \arcsin \delta - b_\delta.$$

This affine angle coordinate maps $[1, \rho]$ exactly to the promised angle interval. Strict convexity of $\arcsin$ on $(0, 1)$ gives $b_\delta > \arcsin \delta > \delta$, $a_\delta < 0$, and $b_\delta \leq C_\rho \delta$ for small δ . Let P_n and S_n be the order- n Taylor polynomials, in y , of $F(a_\delta + b_\delta y)/\delta$ and $\sin(a_\delta + b_\delta y)/\delta$, respectively. For $|y| \leq A_0 := \rho + 1$, Taylor's theorem and trigonometric Bernstein give remainders bounded by

$$u_T = \frac{2(C_\rho A_0 T)^{n+1} \delta^n}{(n+1)!}, \quad u_s = \frac{(C_\rho A_0)^{n+1} \delta^n}{(n+1)!}. \quad (26)$$

Here the sine target is retained: replacing it by its linear part would introduce an error of order δ^2 after rescaling, which is unacceptable when $K \ll \delta^2$. Correctness therefore implies

$$\|P_n - S_n\|_{[1, \rho]} \leq K + u_T + u_s.$$

Decrease K_0 so that $0 < a_n \leq 1$. Global nonnegativity yields $P_n(-a_n) \geq -u_T$. Also $\arcsin(\delta a_n) \leq a_n \arcsin \delta$, since $\arcsin x/x$ increases for positive x . Hence

$$b_\delta(1 + a_n) \geq \arcsin \delta + \arcsin(\delta a_n), \quad a_\delta - b_\delta a_n \leq -\arcsin(\delta a_n).$$

All these angles lie in $(-\pi/2, \pi/2)$ for small δ , uniformly in n , so $S_n(-a_n) \leq -a_n + u_s$. The exterior Chebyshev inequality, proved in Theorem 4.2 and classically stated in [2, Proposition 2.1], now gives

$$a_n - u_T - u_s \leq (K + u_T + u_s) T_n(z_n).$$

Since $T_n(z_n) \geq 1$ and $a_n/T_n(z_n) = H_n$,

$$H_n \leq K + 2u_T + 2u_s. \quad (27)$$

Maximality of n gives $8K > H_{n+2} \geq c_\rho(n+2)^{-1} R_0^{-n-2}$, using (23). Consequently

$$\frac{u_s}{K} \leq C_\rho(n+2) \frac{(C_\rho R_0 \delta)^n}{(n+1)!}.$$

For sufficiently small δ_0 this is at most one simultaneously for every allowed n . Equations (27) and $H_n \geq 8K$ then imply $u_T \geq (5/2)K$. Solving (26) and using $((n+1)!)^{1/(n+1)} \geq (n+1)/e$ proves (24). Finally, the two inequalities $H_n \geq 8K > H_{n+2}$ and (23) give (25). The proof uses the periodic oracle family globally, but invokes correctness only on its principal-branch promised interval; it never assumes the output depends solely on $\sin t$. $\square$

5.2 A contractive upper bound and a matched precision law

Proposition 5.2 (Integrated sign construction). *For $0 < \delta < 1$ and $0 < \eta < 1/4$, there is a real polynomial q of degree $O(\delta^{-1} \log(1/\eta))$ with $0 \leq q \leq 1$ on $[-1, 1]$ and*

$$|q(x) - (1 - x)| \leq \eta \quad (\delta \leq x \leq 1).$$

It gives a unit-normalized converter with the same query order.

Proof. The bounded sign-approximation lemma [9, Lemma 25, extended version] supplies an odd real polynomial s of degree $O(\delta^{-1} \log(1/\eta))$, bounded by one on $[-1, 1]$, with $|s(x) - \text{sign } x| \leq 2\eta$ for $|x| \geq \delta$. Set

$$p(x) = \int_{-1}^x \frac{1 + s(t)}{2} dt, \quad q(x) = 1 - p(x).$$

Oddness gives $p(1) = 1$, and $0 \leq p' \leq 1$ gives $0 \leq p, q \leq 1$. For $x \geq \delta$,

$$q(x) - (1 - x) = \frac{1}{2} \int_x^1 (s(t) - 1) dt,$$

whose absolute value is at most η . Apply Lemma 2.1; mixed parity incurs no normalization loss. $\square$

Theorem 5.3 (Matched high-accuracy law). *For each fixed $\beta > 0$, uniformly for $0 < K \leq \delta^\beta$ and sufficiently small δ satisfying (1),*

$$Q(\delta, K\delta; \rho, c) = \Theta_{\rho, \beta}(\delta^{-1} \log(1/K)). \quad (28)$$

Let $Q_{\text{comp}}(\delta, \eta)$ instead denote the worst-case query complexity for all positive contractions $\delta I \preceq H \preceq I$, with absolute complement error η . For every fixed $\beta > 0$, uniformly for $0 < \eta \leq \delta^{1+\beta}$,

$$Q_{\text{comp}}(\delta, \eta) = \Theta_\beta(\delta^{-1} \log(1/\eta)). \quad (29)$$

Proof. Equation (24), with (25), is at least $c_\rho \delta^{-1} L \exp(-C_\rho D/L)$. If $L \geq \beta D$, the exponential factor is bounded below by a positive constant depending only on ρ, β . Proposition 5.2 gives the upper bound $O(\delta^{-1}(D + L))$, which is $O_\beta(\delta^{-1} L)$. For the full-interval problem, take $\rho = 2$ for the lower bound and $K = \eta/\delta$. Writing $E = \log(1/\eta)$ gives $L = E - D \geq \beta D$, hence $L \geq \beta E/(1 + \beta)$. The upper construction is already accurate on $[\delta, 1]$. $\square$

These conclusions concern reusable coherent output. They apply regardless of matrix dimension or sparsity when the scalar continuum is included in the access contract. They need not hold for known finite spectra (Section 3) or direct access to the normalized complement. Standard uniform singular value amplification [9, Theorem 30, extended version] also gives the upper order after first encoding $(I - H)/2$; the new lower-bound issue is the continuum obstruction under unit normalization.

5.3 An independent nonnegative-square lower bound

We next use Fejér–Riesz factorization instead of exterior extrapolation of a Taylor polynomial. In particular, this route does not use Theorem 5.1.

Lemma 5.4 (Factor and analytic Taylor remainder). *For a T -query converter put $F(t) = 1 - \operatorname{Re} q(t)$. There is a polynomial A of degree at most T such that*

$$F(t) = |A(e^{it})|^2, \quad \|A\|_{|z|=1} \leq \sqrt{2}.$$

Let $b(x) = A(e^{i \arcsin x})$, analytic for $|x| < 1$, and let B_r be its Taylor polynomial of degree r . Set $s = r + 1$. For $\rho\delta < R \leq 1/4$,

$$e_r := \sup_{1 \leq y \leq \rho} |b(\delta y) - B_r(\delta y)| \leq \frac{\sqrt{2}e^{2TR}}{1 - \rho\delta/R} (\rho\delta/R)^s. \quad (30)$$

If the converter has error $K\delta$ on the low interval, then, with $u = e_r/\sqrt{\delta}$,

$$G_r \leq K + 2\sqrt{\rho + K}u + u^2. \quad (31)$$

Proof. The first assertion is the classical Fejér–Riesz theorem; its root-factorization proof is also given in [15, proof of Theorem 4]. To recall why it applies, multiply the nonnegative Laurent polynomial F of actual degree d by z^d . Its roots away from the unit circle occur in conjugate-reciprocal pairs, and its roots on the circle have even multiplicity, because a real analytic nonnegative function has zeros of even order. Selecting one root from each reciprocal pair and half of each circle multiplicity gives A ; positivity fixes a nonnegative scalar factor. The case $F \equiv 0$ is immediate. Its circle norm follows from $F \leq 2$. The maximum principle applied to A inside the unit disk and to $z^{-T}A(z)$ outside it gives

$$|A(z)| \leq \sqrt{2} \max(1, |z|^T).$$

The power series for $\arcsin$ has nonnegative coefficients and yields $|\arcsin z| \leq \arcsin R \leq 2R$ for $|z| = R \leq 1/4$. Thus $|b(z)| \leq \sqrt{2}e^{2TR}$. Cauchy’s coefficient estimate and summation of a geometric tail prove (30). For real y , the polynomial

$$P(y) = \delta^{-1} B_r(\delta y) \overline{B_r(\delta y)}$$

(where conjugation acts on coefficients) is real, globally nonnegative, and of degree at most $2r$. On the low interval, $|b(\delta y)| \leq \sqrt{(\rho + K)\delta}$ and $||b(\delta y)|^2/\delta - y| \leq K$. The inequality $||B|^2 - |b|^2| \leq 2|b||B - b| + |B - b|^2$ and the definition of G_r prove (31). $\square$

Theorem 5.5 (Uniform finite-index margin bound). *There are $c_\rho, C_\rho > 0$ such that, for $0 < \delta \leq 1/(2\rho)$, $r \geq 1$, $K < G_r$, and $r \leq c_\rho \delta^{-1/2}$, every converter accurate to $K\delta$ on the low interval satisfies*

$$T \geq C_\rho(r+1)(G_r - K)^{1/(r+1)} \delta^{-1+1/(2r+2)}. \quad (32)$$

Proof. First, $G_r \leq G_1 < G_0 = (\rho - 1)/2$. The endpoint magnitudes of the factor in Lemma 5.4 differ by at least

$$(\sqrt{\rho - K} - \sqrt{1 + K})\sqrt{\delta} \geq d_\rho \sqrt{\delta},$$

with $d_\rho > 0$ since $K < G_1$. Bernstein’s inequality for the analytic trigonometric polynomial $A(e^{it})$ bounds its derivative by $\sqrt{2}T$. The two endpoint angles differ by at most $C_\rho \delta$, so $T \geq d'_\rho \delta^{-1/2}$. Decreasing c_ρ ensures $s = r + 1 \leq T$. If $T\rho\delta \geq s/8$, the resulting $T \geq s/(8\rho\delta)$ is stronger than (32), because $(G_r - K)^{1/s} \delta^{1/(2s)}$ is bounded above by a ρ -dependent constant. Otherwise use $R = s/(4T) \leq 1/4$, so $\rho\delta/R < 1/2$ in (30). It gives

$$e_r \leq 2\sqrt{2} (4\sqrt{e} T \rho \delta / s)^s. \quad (33)$$

By (31),

$$u \geq \sqrt{\rho + G_r} - \sqrt{\rho + K} = \frac{G_r - K}{\sqrt{\rho + G_r} + \sqrt{\rho + K}} \geq c_\rho(G_r - K).$$

Combine this with (33) and take an s th root. Constants remain uniform because $s \geq 2$. $\square$

Theorem 5.6 (All-index bound from a fixed-radius branch). *For sufficiently small $\delta > 0$, depending only on ρ , every integer $r \geq 1$ satisfying*

$$K \leq \frac{\sqrt{\rho}}{4er} R_0^{-2r} \quad (34)$$

gives the lower bound

$$T \geq c_\rho r \delta^{-1+1/(2r+2)}. \quad (35)$$

There is no upper restriction on r or on $\log(1/K)$.

Proof. Evaluating (8) at $z = \cosh(a + 1/(2r))$, using convexity of $\cosh$ and $\cosh(2ra + 1) \leq e^{2ra+1}$, gives the explicit estimate

$$G_r \geq \frac{\sqrt{\rho}}{2er} R_0^{-2r}. \quad (36)$$

Thus (34) implies $G_r - K \geq G_r/2$. The obstruction in Lemma 5.4 consequently requires

$$e_r \geq c_\rho \sqrt{\delta} r^{-1} R_0^{-2r}. \quad (37)$$

If $T < s = r + 1$, choose the fixed radius $R = 1/4$. For $\rho\delta \leq 1/8$, (30) gives

$$e_r \leq 2\sqrt{2}e^{s/2}(4\rho\delta)^s.$$

The ratio of this upper bound to the right side of (37) is at most

$$C_\rho r \delta^{r+1/2} (4\rho\sqrt{e} R_0^2)^r.$$

It is less than one for every $r \geq 1$ simultaneously when δ is sufficiently small: choose $C'_\rho \delta < 1/2$ and use $\sup_{r \geq 1} r 2^{-r} \leq 1$. Hence $T \geq s$. The adaptive-radius argument of Theorem 5.5 now applies without its auxiliary restriction on r . Taking an s th root of (36) costs only a fixed factor, since $r^{-1/(r+1)}$ is bounded below and $R_0^{-2r/(r+1)} \geq R_0^{-2}$. This proves (35). $\square$

In particular, for $0 < K \leq \sqrt{\rho} R_0^{-2}/(4e)$, the following maximum has a finite, nonempty index set and gives a completely explicit joint lower bound:

$$T \geq c_\rho \max_{r \geq 1: K \leq \sqrt{\rho} R_0^{-2r}/(4er)} r \delta^{-1+1/(2r+2)}. \quad (38)$$

For large L , the largest admissible integer is $r = (L - \log L + O_\rho(1))/(2a)$, obtained by taking logarithms in (34). Alternatively, choosing $r = \lfloor \theta_\rho L \rfloor$ for any fixed sufficiently small $\theta_\rho > 0$ gives the unambiguous uniform consequence

$$T \geq c_\rho \frac{L}{\delta} \exp(-C_\rho D/L). \quad (39)$$

Together with Proposition 5.2, this independently proves Theorem 5.3. It also covers $L/D \rightarrow \infty$ with no growth restriction. If $K = 0$, every r satisfies (34); letting $r \rightarrow \infty$ in (35) rules out any finite-query exact converter on the low continuum, for sufficiently small δ .

5.4 A pinned construction with a growing threshold index

The next theorem controls all constants in the pinned construction, including constants raised to the degree.

Theorem 5.7 (Uniform growing-index upper bound). *Fix $\rho > 1$ and $0 < c \leq 1$. Suppose $K = K(\delta) \downarrow 0$ and*

$$r = \min\{j \geq 1 : G_j \leq K\} \longrightarrow \infty, \quad r = o(D).$$

Then, for all sufficiently small δ along this parameter choice,

$$Q(\delta, K\delta; \rho, c) \leq C_{\rho, c} \left(r^3 \delta^{-1+1/(2r)} + D \right). \quad (40)$$

The constant is independent of the growing r . The construction has low-band error at most exactly $G_r \delta$ and high-band error at most $K\delta$.

Proof. Put $n = 2r$, $p = r + 2$, $s = r + 1$, and $P = P_r^*$ from (9). The two-sided bounds

$$c_\rho r^{-1} R_0^{-n} \leq G_r \leq C_\rho r^{-1} R_0^{-n} \quad (41)$$

hold for every $r \geq 1$ by Proposition 4.3 and adjustment of finitely many constants. Its positive-error contacts are

$$y_j = m_\rho + h_\rho \cos(2\pi j/n), \quad 0 \leq j \leq r.$$

Let k be the least odd integer at least $A\delta^{-1+1/n}$, where the fixed $A = A_\rho$ will be chosen below. With $a_j = \arcsin(\delta y_j)$ and $\theta = \arcsin x$, define

$$W(x) = 1 - \prod_{j=0}^r \prod_{\sigma=\pm 1} \left(1 - D_{k, \sigma a_j}(\theta)^{2p} \right)^s, \quad D_{k, b}(\theta) = \frac{\sin(k(\theta - b))}{k \sin(\theta - b)}. \quad (42)$$

The same argument as Lemma 4.5 proves that W is an even polynomial, $0 \leq W \leq 1$, and $W(\delta y_j) = 1$. More explicitly, its trigonometric degree and therefore its degree in x are at most $4(r+1)^2(r+2)(k-1)$. Notice

$$k\delta \leq 2A\delta^{1/n} = o(1), \quad k \geq A\delta^{-1+1/n}, \quad (43)$$

for sufficiently small δ . These inequalities suffice; no degree-dependent constant is hidden in integer rounding.

For the low interval write $y = m_\rho + h_\rho \cos \phi$, $0 \leq \phi \leq \pi$, and select a nearest angle $\phi_j = 2\pi j/n$. Then $\Delta = |\phi - \phi_j| \leq \pi/n$, and $d = |y - y_j| \leq h_\rho \Delta \leq C_\rho/r$. The normalized Dirichlet kernel is the average of the exponentials with frequencies $-(k-1), -(k-3), \dots, k-1$. Therefore $1 - D_{k,0}(u) \leq k^2 u^2/2$ and, since $|D| \leq 1$, $1 - D_{k,0}(u)^{2p} \leq p(1 - D_{k,0}(u)^2) \leq pk^2 u^2$. Using just the nearest positive-contact factor in $1 - W$ gives

$$\lambda(y) := 1 - W(\delta y) \leq [C_\rho r (k\delta)^2 d^2]^{r+1}. \quad (44)$$

On the other hand, with $e(y) = P(y) - y$,

$$G_r - e(y) = G_r [1 - \cos(n\Delta)] \geq \frac{2G_r n^2 \Delta^2}{\pi^2} \geq c_\rho G_r r^2 d^2. \quad (45)$$

Dividing (44) by δ times this slack, using (41), (43), and $d \leq C_\rho/r$, yields

$$\frac{\lambda(y)}{\delta(G_r - e(y))} \leq C_{\rho, A} e^{-D/r} \left(\frac{C_{\rho, A} R_0^2}{r} \right)^r = o(1). \quad (46)$$

At contacts the ratio is interpreted by continuity; both numerator and denominator vanish and the displayed estimate remains valid. In particular $W = 1 - o(1)$ uniformly on the low interval, and eventually $\lambda \leq \frac{1}{2}W\delta(G_r - e)$.

Choose S_N as in (13), and set

$$q(x) = W(x)[1 - \delta P(x/\delta)] + (1 - W(x))S_N(x^2). \quad (47)$$

For $x = \delta y$ its signed error is $(1 - x) - q(x) = W\delta e(y) + \lambda d_N(x)$, where $d_N(x) = 1 - x - S_N(x^2) \in [0, 1]$. The lower error bound is $-G_r\delta$. The upper bound follows from

$$G_r\delta - [(1 - x) - q(x)] = W\delta(G_r - e) + \lambda(G_r\delta - d_N) \geq 0.$$

This proves the exact low-band threshold error.

We next verify global contractivity, with explicit tail constants. For $|x| \geq 4\rho\delta$ and small δ ,

$$|\sin(\arcsin x \pm a_j)| \geq |x|\sqrt{1 - \rho^2\delta^2} - \rho\delta \geq |x|/2.$$

The elementary product bound therefore gives

$$W(x) \leq \min \left\{ 1, 2(r+1)^2 \left(\frac{2}{k|x|} \right)^{n+4} \right\}. \quad (48)$$

The factor 2^{n+4} is essential and cannot be absorbed into a constant independent of r . For a fixed $B_\rho \geq 1$, the elementary Chebyshev estimate $|T_n(v)| \leq [2(1 + |v|)]^n$ gives

$$0 \leq P(y) \leq |y| + G_r[B_\rho(1 + |y|)]^n. \quad (49)$$

On $|x| \leq 4\rho\delta$, this bounds $\delta P(x/\delta)$ by $4\rho\delta + C_\rho\delta\Lambda_\rho^n/r = o(1)$, since $n = o(D)$. On $4\rho\delta \leq |x| \leq 1/k$, use $W \leq 1$ and enlarge a fixed B_ρ to replace $1 + |x|/\delta$ by a constant times $|x|/\delta$. Then

$$\delta P(x/\delta)W(x) \leq k^{-1} + \frac{C_\rho}{r} \left(\frac{B_\rho}{AR_0} \right)^n.$$

Here $\delta(k\delta)^{-n} \leq A^{-n}$ by the lower bound in (43). Finally, on $|x| \geq 1/k$, which exceeds $4\rho\delta$ eventually, combine (48) and (49). The linear term is at most $Cr^22^n/k = o(1)$ because its logarithm is at most $-D + D/n + O_\rho(n + \log r)$. The degree- n term decreases as $|x|^{-4}$, and at $|x| = 1/k$ is at most

$$C_\rho r \left(\frac{2B_\rho}{AR_0} \right)^n.$$

Choose A large enough that $2B_\rho/(AR_0) < 1/2$. All three regions then give $\sup_{[-1,1]} \delta P(x/\delta)W(x) = o(1)$, in particular at most one eventually. In (47) write $q = B - \delta PW$, with $B = W + (1 - W)S_N \in [0, 1]$. Since $P \geq 0$, this proves $-1 \leq q \leq 1$ everywhere.

For the high interval $[c, 1]$, (48) implies $W \leq Cr^2(2/(kc))^{n+4}$. Separating the linear and degree- n terms of (49) gives

$$\begin{aligned} \sup_{[c,1]} \frac{W(1 + |x|)}{G_r\delta} &\leq C_{\rho,c}r^3 \left(\frac{C_{\rho,c}R_0}{A} \right)^n \delta^{n+2-4/n} = o(1), \\ \sup_{[c,1]} \frac{WG_r\delta[B_\rho(1 + |x|/\delta)]^n}{G_r\delta} &\leq C_{\rho,c}r^2 \left(\frac{C_{\rho,c}}{A} \right)^n \delta^{3-4/n} = o(1). \end{aligned}$$

Fixed factors A^{-4} have been bounded by one. Even if a displayed base exceeds one, its logarithm is $O_{\rho,c}(n) = o(D)$, whereas the last power of δ has logarithm $-(3 - o(1))D$. Thus the gate contribution $W|1 - \delta P - S_N|$ is $o(G_r \delta)$, including the term W itself. Choose the positive-series tail at most $K\delta/2$: for $c < 1$ it suffices to take

$$N + 1 \geq \frac{\log(2/(K\delta))}{-\log(1 - c^2)},$$

and for $c = 1$ take $N = 1$. Since $G_r \leq K$, (41) gives $L \leq na + \log r + O_\rho(1) = O_\rho(r + \log r) = o(D)$. Hence $N = O_c(D)$. The high error is at most $K\delta$ eventually, and the degree of (47) is $O_\rho(r^3 k + N)$. Lemma 2.1 completes the proof. $\square$

5.5 What the joint bounds determine

Combining the finite-margin lower bound with the growing-index upper bound gives a sharper comparison than treating the two estimates separately.

Corollary 5.8 (Intermediate accuracy away from a tier boundary). *Under the assumptions of Theorem 5.7, for all sufficiently small δ ,*

$$c_\rho r (G_{r-1} - K)^{1/r} \delta^{-1+1/(2r)} \leq Q(\delta, K\delta; \rho, c) \leq C_{\rho,c} r^3 \delta^{-1+1/(2r)}. \quad (50)$$

For each fixed $\gamma \in (0, 1)$, if also $K \leq (1 - \gamma)G_{r-1}$, then

$$c_{\rho,\gamma} r \delta^{-1+1/(2r)} \leq Q(\delta, K\delta; \rho, c) \leq C_{\rho,c} r^3 \delta^{-1+1/(2r)}. \quad (51)$$

In particular the latter comparison holds at the exact lower thresholds $K = G_r$, with a fixed γ depending only on ρ , for all large r .

Proof. Since $r = o(D)$, eventually $r - 1 \leq c_\rho \delta^{-1/2}$. Minimality of r gives $K < G_{r-1}$, so Theorem 5.5 at index $r - 1$ proves the lower half of (50). Theorem 5.7 proves the upper half; its additive D is absorbed since $r \geq 2$ eventually and $D = o(\delta^{-3/4})$. If the stated margin holds, (41) implies

$$(G_{r-1} - K)^{1/r} \geq \left(\frac{c_\rho \gamma}{r - 1} R_0^{-2r+2} \right)^{1/r} \geq c_{\rho,\gamma} > 0.$$

Finally $G_r/G_{r-1} \rightarrow R_0^{-2} < 1$, so any fixed $\gamma < (1 - R_0^{-2})$ applies at $K = G_r$ eventually. $\square$

For $K \rightarrow 0$ with $L = o(D)$, Proposition 4.3 gives $r = \Theta_\rho(L) = o(D)$. Equations (39) and (40) then imply

$$\frac{\log Q(\delta, K\delta; \rho, c)}{\log(1/\delta)} \rightarrow 1.$$

Thus the exponent tends to one throughout this intermediate regime. Away from the upper boundary of a tier, (51) narrows the multiplicative gap to $O_{\rho,c,\gamma}(r^2) = O_{\rho,c,\gamma}(L^2)$, including exact threshold accuracy. Near that upper boundary, the explicit factor $(G_{r-1} - K)^{1/r}$ in (50) must be retained; the independent bound (38) remains available. We do not claim an optimal uniform multiplicative law in this intermediate regime. Further questions include determining the remaining multiplicative factor and sharply controlling approach to the upper tier boundary. In contrast, Theorem 5.3 determines the full order whenever L is at least a fixed positive multiple of D .

6 An exact-central sparse LP and its oracle separation

We realize the spectral family in an actual primal–dual Newton system. The state task below outputs the normalized dual Newton direction; it allows measurements and discards auxiliary registers. The compiler task instead outputs a reusable coherent block encoding. Their input contracts are stated separately.

6.1 The linear program and the predictor equation

Fix $\rho > 1$ and a public $\delta > 0$ so small that $\delta < 1/8$ and $\rho\delta < 1/4$. The parameter $t \in [\delta, \rho\delta]$ is hidden. Define

$$\begin{aligned} u_{\pm} &= (1, \pm 1)^T / \sqrt{2}, & q_+ &= (1, 1, 2)^T / \sqrt{6}, \\ v_0 &= (1, -1, 0)^T / \sqrt{2}, & k &= (1, 1, -1)^T / \sqrt{3}, \\ \sin \theta &= 2\sqrt{2\delta}, \quad 0 < \theta < \pi/2, & q_- &= \cos \theta v_0 + \sin \theta k, \\ & & q_0 &= -\sin \theta v_0 + \cos \theta k. \end{aligned}$$

The triples q_+, q_-, q_0 and the pair u_+, u_- are orthonormal. Put

$$A_t = u_+ q_+^T + \sqrt{t} u_- q_-^T, \quad H_t = A_t A_t^T = P_+ + t P_-, \quad P_{\pm} = u_{\pm} u_{\pm}^T. \quad (52)$$

Thus $\|A_t\| = \|H_t\| = 1$, A_t has rank two, and $\kappa(H_t) = 1/t = \Theta_{\rho}(\delta^{-1})$. Every row of A_t has at most three nonzero entries and every column at most two; its entries have absolute value at most one. Sparsity here is a constant degree property of this fixed-size example, not an asymptotic dimension claim.

Proposition 6.1 (A genuine central Newton system). *The three-variable, two-equality linear program*

$$\min \mathbf{1}^T x \quad \text{subject to} \quad A_t x = b_t := A_t \mathbf{1}, \quad x \geq 0 \quad (53)$$

has a nontrivial compact feasible segment and a nonconstant objective on that segment. The point $x = s = \mathbf{1}$, $y = 0$ is exactly central with barrier parameter $\mu = 1$. Its affine-scaling predictor has normal equation and solution

$$H_t \Delta y = b_t = a(u_+ + \sqrt{\delta t} u_-), \quad a = 2\sqrt{2/3}, \quad (54)$$

$$\Delta y = a(u_+ + \sqrt{\delta/t} u_-). \quad (55)$$

Proof. The feasible affine line is $\mathbf{1} + \lambda q_0$. Since q_0 is nonzero and orthogonal to the strictly positive vector q_+ , it has both positive and negative entries. The inequalities $1 + \lambda(q_0)_j \geq 0$ therefore bound λ on both sides and admit an open neighborhood of zero. Their intersection is a nontrivial compact interval. Moreover,

$$\mathbf{1}^T q_0 = \cos \theta / \sqrt{3} > 0,$$

so the objective varies strictly along this line. For the dual convention $A_t^T y + s = \mathbf{1}$, centrality is the system $A_t x = b_t$, $A_t^T y + s = \mathbf{1}$, $x_j s_j = \mu$, $x, s > 0$. The stated point satisfies it at $\mu = 1$. Linearizing with target complementarity zero gives exactly

$$A_t \Delta x = 0, \quad A_t^T \Delta y + \Delta s = 0, \quad \Delta x + \Delta s = -\mathbf{1}.$$

Eliminating $\Delta x, \Delta s$ yields $H_t \Delta y = A_t \mathbf{1}$. Finally, $q_+^T \mathbf{1} = a$ and $q_-^T \mathbf{1} = a\sqrt{\delta}$, proving both formulas. $\square$

The parameter t changes the representation of the equalities, not the feasible segment: the row space of A_t is fixed. In fact $\Delta x = -(\cos \theta / \sqrt{3})q_0$ is independent of t , whereas the dual direction (55) is not. This distinction matters for interpreting the state lower bound.

6.2 Complete oracle contracts

All spaces and basis changes below are public, including their dependence on δ . No numerical value of t , A_t , b_t , or their hidden coefficients is supplied as classical data. In particular, the exact norm $\|b_t\| = a\sqrt{1 + \delta t}$ is not supplied: it would reveal t . The objective, central point, formulas, δ , and ρ are public. Each controlled call to a supplied oracle or its inverse costs one query; coherent choices of query type are allowed and counted. Oracle-independent gates and classical computation are free, as in the rest of the paper.

For normal-matrix access supply the following full unitary on $\mathbb{C}^2 \otimes \mathbb{C}^2$, with the first factor the block flag:

$$U_H(t) = \begin{pmatrix} H_t & \sqrt{I - H_t^2} \\ \sqrt{I - H_t^2} & -H_t \end{pmatrix}. \quad (56)$$

Its $|0\rangle$ compression is exactly H_t . In the u_+ and u_- sectors its blocks are respectively $R(1)$ and $R(t)$, where

$$R(z) = \begin{pmatrix} z & \sqrt{1 - z^2} \\ \sqrt{1 - z^2} & -z \end{pmatrix}.$$

For rectangular-factor access let $\mathcal{R} = \mathbb{C}^2$ be the row-index output space and $\mathcal{C} = \mathbb{C}^3$ the column-index input space, and supply the full unitary on $\mathcal{R} \oplus \mathcal{C}$

$$U_A(t) = \begin{pmatrix} \sqrt{I_{\mathcal{R}} - A_t A_t^T} & A_t \\ A_t^T & -\sqrt{I_{\mathcal{C}} - A_t^T A_t} \end{pmatrix}. \quad (57)$$

The column-to-row off-diagonal block is A_t . More precisely, for the canonical embeddings $J_{\mathcal{R}}, J_{\mathcal{C}}$, $J_{\mathcal{R}}^* U_A(t) J_{\mathcal{C}} = A_t$. On each ordered pair (u_j, q_j) this unitary is $\begin{pmatrix} \sqrt{1 - \sigma_j^2} & \sigma_j \\ \sigma_j & -\sqrt{1 - \sigma_j^2} \end{pmatrix}$, where $\sigma_+ = 1$ and $\sigma_- = \sqrt{t}$; it acts as -1 on q_0 . These blocks verify unitarity and specify the completion on the entire space. Fixed identity padding, if needed for a qubit implementation, is public and independent of t .

Both state-access models also supply the right-side preparation unitary B_t on $\mathcal{R}$. In the ordered basis (u_+, u_-) its full matrix is

$$B_t = \frac{1}{\sqrt{1 + \delta t}} \begin{pmatrix} 1 & -\sqrt{\delta t} \\ \sqrt{\delta t} & 1 \end{pmatrix}, \quad B_t u_+ = b_t / \|b_t\|. \quad (58)$$

Calls to B_t and B_t^* , including controlled calls, are counted. Fixing the complete unitaries is essential: unspecified completion blocks could reveal t through an unrelated register.

6.3 Matching query bounds for the Newton state

Write $D(\omega, \tau) = \frac{1}{2} \|\omega - \tau\|_1$ for trace distance and

$$|\psi_t\rangle = \frac{u_+ + \sqrt{\delta/t} u_-}{\sqrt{1 + \delta/t}}, \quad d_\rho = \frac{\sqrt{\rho} - 1}{\sqrt{2(\rho + 1)}}. \quad (59)$$

The task is to output a density operator ω_t with $D(\omega_t, |\psi_t\rangle\langle\psi_t|) \leq \epsilon_0$ for every t in the interval, using a bounded worst-case number of queries. The density operator includes all measurement randomness; success is not defined by conditioning on a favorable event.

Theorem 6.2 (Normal-matrix versus factor state access). *For each fixed $\rho > 1$ and $0 < \epsilon_0 < d_\rho/4$, the optimal worst-case total query counts in the two state-access models are*

$$Q_H^{\text{state}}(\delta, \epsilon_0) = \Theta_{\rho, \epsilon_0}(\delta^{-1}), \quad Q_A^{\text{state}}(\delta, \epsilon_0) = \Theta_{\rho, \epsilon_0}(\delta^{-1/2}). \quad (60)$$

Equivalently, their orders are $\kappa(H_t)$ and $\sqrt{\kappa(H_t)}$, uniformly up to ρ -dependent constants on the family.

Proof. For the lower bounds, the two endpoint target states have distance exactly d_ρ : their overlap is $(\sqrt{\rho} + 1)/\sqrt{2(\rho + 1)}$. Correct output states thus have distance at least $d_\rho - 2\epsilon_0 > d_\rho/2$. For $0 \leq x, y \leq 1/2$,

$$\|R(x) - R(y)\| = 2 \left| \sin \frac{\arcsin x - \arcsin y}{2} \right| \leq 2|x - y|.$$

The factor blocks obey the same bound in their singular-value parameter, by a fixed change of basis. At the endpoints, therefore,

$$\begin{aligned} \|U_H(\rho\delta) - U_H(\delta)\| &\leq 2(\rho - 1)\delta, \\ \|U_A(\rho\delta) - U_A(\delta)\| &\leq 2(\sqrt{\rho} - 1)\sqrt{\delta}. \end{aligned}$$

The preparation unitary is a rotation through angle $\arctan \sqrt{\delta t}$, so

$$\|B_{\rho\delta} - B_\delta\| \leq (\sqrt{\rho} - 1)\delta.$$

These bounds are unchanged for inverses and controlled calls. Purify the algorithm, defer its measurements, and replace queries one at a time between the two endpoint computations. The resulting vector distance is at most the sum of the per-query operator distances. Trace distance cannot increase on discarding registers and is at most this vector distance. Consequently T total queries give output distance at most $C_\rho T\delta$ in the normal model and $C_\rho T\sqrt{\delta}$ in the factor model. Comparing with $d_\rho/2$ proves both lower bounds, including algorithms using the right-side oracle.

For the upper bounds we estimate the single hidden parameter. Standard amplitude estimation [3, Theorem 12], applied to a unitary preparation with success probability a_0 , uses $O(M)$ calls to that preparation and its inverse and returns $\hat{a} \in [0, 1]$ such that

$$|\hat{a} - a_0| \leq \frac{2\pi\sqrt{a_0(1-a_0)}}{M} + \frac{\pi^2}{M^2} \quad (61)$$

with probability at least $8/\pi^2 > 1/2$. Reflections about the public initial states and success subspaces are free; reflecting about a prepared state uses the counted preparation and inverse calls. In the normal model, apply $U_H(t)$ to $|0\rangle u_-$ and declare flag zero successful; then $a_0 = t^2$. Take $M \geq C/\delta$ and define $\hat{t}$ by clipping $\sqrt{\hat{a}}$ to $[\delta, \rho\delta]$. On the event (61), clipping cannot increase its distance to t , and

$$|\hat{t} - t| \leq \frac{|\hat{a} - t^2|}{t} \leq \delta \left(\frac{2\pi}{C} + \frac{\pi^2}{C^2} \right).$$

In the factor model, apply $U_A(t)$ to $J_C q_-$ and declare the output subspace $\mathcal{R}$ successful; then $a_0 = t$. Taking $M \geq C/\sqrt{\delta}$ and clipping $\hat{a}$ instead gives

$$|\hat{t} - t| \leq \delta \left(\frac{2\pi\sqrt{\rho}}{C} + \frac{\pi^2}{C^2} \right)$$

on the same success event. Choose $C = C_{\rho, \epsilon_0}$ so these bounds are at most $\epsilon_0 \delta$. An odd, fixed number of independent repetitions and their median make the failure probability at most $\epsilon_0/2$; the number of repetitions depends only on ϵ_0 .

Given the resulting estimate, prepare $|\psi_t\rangle$ using a rotation in the known basis. Its angle is $\varphi(t) = \arctan \sqrt{\delta/t}$, with

$$|\varphi'(t)| = \frac{\sqrt{\delta}}{2\sqrt{t}(t+\delta)} \leq \frac{1}{4\delta} \quad (t \geq \delta).$$

On successful estimation the pure-state trace error is thus at most $\epsilon_0/4$. Convexity of trace distance bounds the error of the average output by $\epsilon_0/4 + \epsilon_0/2 < \epsilon_0$, including all failure events. These algorithms use no calls to B_t , and use $O_{\rho, \epsilon_0}(\delta^{-1})$ or $O_{\rho, \epsilon_0}(\delta^{-1/2})$ matrix or factor queries, respectively. Rounding M upward to a power of two changes only a constant factor. This proves the matching upper bounds. $\square$

The identity $\Delta y = (A_t^T)^+ \mathbf{1}$ also follows directly from the singular vectors, and the squared overlap of $\mathbf{1}/\sqrt{3}$ with $\text{span}\{q_+, q_-\}$ is $(8/9)(1+\delta)$. It is bounded away from zero. Factor-based improvements and the role of this support overlap are established in Orsucci and Dunjko [16, Sections 1.4 and 5.4–5.5]. Here the explicit estimation argument avoids generic solver overhead and gives the exact orders at fixed output error. It is a state-output algorithm, not a reusable coherent converter.

6.4 The complement compiler on the LP family

Let $Q_H^{\text{LP}}(\delta, \eta)$ be the minimum query count for a single reusable converter that, for every $t \in [\delta, \rho\delta]$, has a designated block within η of

$$I - H_t = (1-t)P_-.$$

For this task the sole input oracle is (56); the preparation oracle B_t is not supplied. All other data remain public. Define Q_A^{LP} analogously with (57) as its sole oracle and with output on $\mathcal{R}$.

Theorem 6.3 (Compiler transfer and factor bypass). *For fixed $K > 0$ and $\ell = \min\{j \geq 0 : G_j(\rho) \leq K\}$, the normal-matrix compiler has*

$$Q_H^{\text{LP}}(\delta, K\delta) = \begin{cases} 0, & \ell = 0, \\ \Theta_{\rho, K, \ell}(\delta^{-1+1/(2^\ell)}), & \ell \geq 1, \end{cases} \quad (62)$$

for sufficiently small δ . Threshold equalities belong to the cheaper tier. For every fixed $\beta > 0$, uniformly for $0 < \eta \leq \delta^{1+\beta}$,

$$Q_H^{\text{LP}}(\delta, \eta) = \Theta_{\rho, \beta}(\delta^{-1} \log(1/\eta)) = \Theta_{\rho, \beta}(\kappa(H_t) \log(1/\eta)). \quad (63)$$

In contrast, two factor queries implement $I - A_t A_t^T$ exactly with normalization one.

Proof. In the public eigenbasis, the normal oracle is the direct sum of a fixed block $R(1)$ and $R(t)$. Replace the latter by $\begin{pmatrix} \sin \tau & \cos \tau \\ \cos \tau & -\sin \tau \end{pmatrix}$ for every real τ . On $\tau = \arcsin t$ it is the actual oracle. Every matrix element of a T -query converter is now a bounded trigonometric polynomial of degree at most T , even if its gates mix the two eigenspaces. Its designated u_- diagonal entry approximates $1 - \sin \tau$ on the promised interval. Thus every lower bound that uses only the low interval in Sections 4–5 applies without change. In particular, Theorem 4.4 gives all positive-index lower bounds, and Theorem 5.1 (or Theorem 5.6) gives the high-accuracy lower bound. No right-side oracle is included in this reduction: its entries would not obey the same degree induction.

The polynomial upper constructions already apply to H_t with the singleton high band $c = 1$. They prove the upper half of every positive tier and (63), using $K = \eta/\delta$ and $\log(\delta/\eta) = \Theta_\beta(\log(1/\eta))$ in the stated regime. For the coarse tier the public operator $(1 - m_\rho\delta)P_-$ is a contraction and differs from $(1 - t)P_-$ by at most $G_0\delta$. Its standard unitary dilation uses no oracle queries. The general $c = 1$ problem of Theorem 4.1 has a constant nonzero query cost because it does not supply the spectral projectors; the present fixed-eigenspace family permits this additional zero-query shortcut. There is no positive-width high interval in this example.

For factor access the real even polynomial $p(s) = 1 - s^2$ has degree two and lies in $[0, 1]$ on $[-1, 1]$. The even, left singular-vector version of QSVT [9, Corollary 18, extended version], applied with the public row and column projectors in (57), therefore gives exactly

$$\sum_{j=\pm} (1 - \sigma_j^2) u_j u_j^T = I_{\mathcal{R}} - A_t A_t^T$$

as a unit-normalized block using two queries. The output acts on the two-dimensional left singular-vector space $\mathcal{R}$; the extra column null vector does not create an extra output eigenvalue. Equivalently, if $\Pi_{\mathcal{C}} = J_{\mathcal{C}} J_{\mathcal{C}}^*$, then

$$J_{\mathcal{R}}^* U_A(t) (I - \Pi_{\mathcal{C}}) U_A(t)^* J_{\mathcal{R}} = I_{\mathcal{R}} - A_t A_t^T.$$

A free block encoding of the public projector $I - \Pi_{\mathcal{C}}$ makes this an explicit two-query construction. $\square$

The same scalar reduction transfers the finite-index and growing-index bounds of Section 5, including their stated intermediate-accuracy limitations. In particular, error $o(\delta)$ implies $\Omega_{\rho,\varepsilon}(\delta^{-1+\varepsilon})$ queries for every fixed $\varepsilon > 0$, while zero error is impossible with finitely many normal-oracle queries on the continuum. If the particular t were public instead, Proposition 3.1 would give a two-query exact conversion for the known spectrum $\{t, 1\}$ even with unknown eigenspaces; the present public eigenspaces would permit direct synthesis with zero queries. The uniform hidden-parameter contract is therefore essential.

6.5 Access assumptions and scope

The separation disappears under direct digital entry access. For example,

$$(H_t)_{11} = (1 + t)/2, \quad (A_t)_{11} = \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{6}} + \sqrt{t}(q_-)_1 \right), \quad (q_-)_1 = \frac{\cos \theta}{\sqrt{2}} + \frac{\sin \theta}{\sqrt{3}} > 0.$$

A single exact value query to either displayed entry reveals t . A digital query with sufficient precision similarly gives a constant-query approximation at fixed state error, with precision and arithmetic costs accounted for separately. Since the dimension and eigenbases are known, that value permits direct state preparation or synthesis of a complement dilation. This is consistent with the digital sparse-access constructions of Orsucci and Dunjko [16, Section 4.3]; indeed H_t is strictly diagonally dominant, with diagonal $(1 + t)/2$ and off-diagonal $(1 - t)/2$.

There is also a one-query coherent linear-combination construction of a normalization-two encoding: compressing a control qubit prepared and unprepared in $|+\rangle$ around $\text{diag}(I, -U_H(t))$ gives $(I - H_t)/2$. Requiring normalization exactly one is what excludes that shortcut. Orsucci and Dunjko already identify the construction of a normalized encoding of $I - \eta A$ as a substantive input requirement [16, Section 4.3]; the present continuum lower bounds quantify that conversion task in the plain matrix-oracle model.

The conclusions are query separations for these specified analog oracles and output tasks. They neither lower-bound all algorithms for the LP nor imply an end-to-end lower bound for quantum interior-point methods. In particular, the primal predictor here is public, a solver need not compile

a unit-normalized complement, and classical runtime or digital precision is not measured by the oracle query count. The contribution of the example is the explicit exact-central LP realization, the matched state-access comparison with counted right-side access, and the transfer of the complement hierarchy with its fixed-eigenspace coarse-tier correction.

7 Conclusion

Unit normalization makes complement conversion sensitive to spectral support and requested accuracy. Known two-point spectra can admit exact constant-query conversion, while a low continuum forces a sequence of optimal query powers determined by globally nonnegative polynomial approximation. The Chebyshev thresholds and pinned constructions determine every fixed-relative-accuracy tier, including its boundary. The even nonnegative constraint changes this staircase: the exact plateau $F_1 = F_2$ produces a concrete exponent-5/6 cost where unrestricted conversion has exponent 1/2. At polynomial and higher relative accuracy, the coherent conversion cost is instead $\Theta(\delta^{-1} \log(1/K))$.

The main quantitative question left open is the multiplicative order in the intermediate joint regime. Away from upper tier boundaries the proved gap is at most quadratic in the threshold index; approaching those boundaries also requires control of the margin factor. Explicit formulas for higher even thresholds F_r would further clarify the parity comparison. Another separate problem is to account for the precision and classical computation needed to synthesize the threshold-attaining circuits, particularly when no error margin is available.

The exact-central LP demonstrates why the input and output contracts must be kept explicit. A dual Newton state, a reusable complement block, and a digital matrix description are different resources even for a fixed sparse instance family. The separation here quantifies conversion from specified analog matrix access; deciding when a larger algorithm can avoid that conversion, or obtain stronger access efficiently, requires its own analysis.

A Reproducibility

The source archive contains the complete L^AT_EX manuscript, bibliography and generated bibliography file, required PDF figure, build instructions, and Python scripts. Building the paper with `make` requires `pdflatex` and `bibtex`, but does not require Python. The target `make figures` regenerates Figure 1 and supplementary threshold and pinned-kernel diagnostics using NumPy, SciPy, and Matplotlib. The scripts use paths relative to their own locations and require no external data or network access.

The threshold computation solves the unique stationary equation in Proposition 4.3 in the offset variable $s = 2r(t - \log R_0)$, avoiding subtraction of nearby roots. It exports the computed thresholds and the comparison with their leading asymptotic to CSV files. The overview checks the closed formula for G_1 and the inequalities needed for its displayed parity range. The supplementary diagnostic evaluates the signed-error identity of the pinned construction near its contacts, avoiding subtraction of nearly equal quantities. These floating-point calculations illustrate the proved formulas; sampled errors are not certificates of global nonnegativity or contractivity. The archive’s README states the software versions used for reproduction and the individual script commands.

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